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SIDP Technical Architecture

Client: Australian Spatial Industry Business Association

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Table of Contents

1	ASSUMPTIONS	5
2	GLOSSARY	6
2.1	ACRONYMS	6
2.2	TERMINOLOGY	10
3	INTRODUCTION	15
3.1	PURPOSE OF THIS DOCUMENT	15
3.2	RELATIONSHIP TO OGC-A NOTIONAL ARCHITECTURE	15
4	GENERAL IMPLEMENTATION PATTERNS	16
4.1	APPLICATION BOUNDARY CONCEPT	16
	FIGURE 1 SERVICE OVERVIEW AND APPLICATION BOUNDARY	16
4.2	WEB SERVICE INTERFACES	17
4.3	OGC WEB SERVICES	18
4.4	STANDARD INTERFACES FOR SIDP	19
	TABLE 1. OWS INTERFACES AND VERSIONS REQUIRED FOR SIDP ARCHITECTURE	19
4.4.1	Notes:	21
4.5	ADDITIONAL INTERFACES FOR SIDP	21
	TABLE 2. PROVISIONAL SERVICE INTERFACES – PROPOSED BUT NOT MANDATED FOR SIDP	22
5	EVENT DIAGRAMS	23
5.1	NOTATION	23
5.2	GRANULARITY	23
6	SIDP APPLICATION PROFILE	42
6.1	THE ROLE OF CONTENT STANDARDS	42
6.1.1	<i>Semantic equivalence</i>	42
6.1.2	<i>Standard styling</i>	42
6.1.3	<i>Service configuration</i>	42
6.2	SPECIFIED CONTENT STANDARDS FOR SIDP	43
	TABLE 3. CONTENT STANDARDS AND APPLICATION PROFILES TO BE USED IN SIDP SCENARIOS	44
6.3	CATALOGUE APPLICATION PROFILES	48
6.3.1	<i>Datasets and services</i>	48
6.3.2	<i>Additional resource types</i>	48
7	DISCUSSION	49
7.1	REPOSITORIES & LIBRARIES	49
7.2	REGISTRY SERVICE	50
7.3	SETUP AND INITIALISATION	51
7.4	PUBLISHING MAPS	51
7.5	MAP PORTRAYAL AND STYLING	52
7.6	CONTENT CREATION	54
7.7	NON-GRAPHICAL DATA	54
7.8	WMS vs WFS	54

7.9	AUTHENTICATION AND SECURITY	55
TABLE 4.	AUTHENTICATION INTERFACES	55
8	BIBLIOGRAPHY	58

1 Assumptions

1. the proposed architecture will be described in terms of the required services identified from analysis of the scenarios provided by SIDP [17]
2. the services will be described in terms of the interface specification to which they must conform, together with any required content standards
3. client functionality is not described in detail. This is largely “standard” mapping client and portal functionality, as found in most rich client and portal generation software. The key for the SIDP architecture is that it is able to bind to the service interfaces specified.
4. as far as possible components will be coupled using interfaces conforming to published open standards; open interfaces will be specified even for transactions taking place wholly within one participating organisation
5. the actual architecture will need to be further refined on the basis of actual components made available as contributions proposed by SIDP participants in response to the RFP
6. existing content standards will be recommended wherever possible, providing they conform to the required interface specifications
7. where new content standards are required, they are clearly identified

2 Glossary

The glossary from the OGC-A Notional Architecture [14] is reproduced here, augmented with additional terms used in this document.

2.1 Acronyms

AAA	Authentication, authorization and accounting
ACC	Australian Corporate Content
ACXML	Access Control Extensible Mark-up Language (OASIS): see http://www.oasis-open.org/committees/tc_home.php?wg_abbrev=xacml
AGI	Association for Geographic Information (UK): see http://www.gigateway.org/
AGLS	Australian Government Locator System
ALGA	Australian Local Government Association: see http://www.alga.com.au/
ANZLIC	Australian New Zealand Land Information Council: see http://www.anzlic.org.au/
ASAP	Aligning Services and Priorities (Government of Qld)
ASDD	Australian Spatial Data Directory: see http://asdd.ga.gov.au/asdd/
ASDI	Australian Spatial Data Infrastructure
ASIBA	Australian Spatial Industry Business Association: see http://www.asiba.com.au/
AUSLIG	Australian Survey and Land Information Group
BOM	Bureau of Meteorology
BON	Business Object Notation: see http://archive.eiffel.com/products/bon.html
BPR	Business Process Review (Queensland State Government)
CANRI	Community Access to Natural Resource Information (NSW) see http://www.canri.nsw.gov.au/
CFA	Victorian Country Fire Authority
CMS	Cascading Map Server (OGC WMS)
CP	Control Point
CQL	Common Catalogue Query Language (Library of Congress, USA): See http://www.loc.gov/z3950/agency/zing/cql/
CRC	Cooperative Research Centre (Australia): see https://www.crc.gov.au/
CSDGM	Content Standard for Digital Geospatial Metadata (USA FGDC)
CS-W	Catalog Service – Web interface (OGC)
CS-W-T	Transactional Catalog Service – Web interface (OGC)

DACS	Distributed Access Control System (OGC): see http://dacs.dss.bc.ca/arch-spec.html
DES	Department of Emergency Services (Qld)
DIIE	Department of Innovation and the Information Economy (Qld): see http://www.iie.qld.gov.au/
DLGP	Department of Local Government and Planning (Qld): see http://www.dlgp.qld.gov.au/
DIL	Department of Land Information (WA)
DOL	Department of Lands New South Wales
EbRIM	ebXML registry information model (OASIS) see http://www.ebxml.org/specs/ebRIM.pdf
EbXML	E-business Extensible Mark-up Language (OASIS): see http://www.ebxml.org/
EC	European Commission
e-GIF	E-Government Interoperability Framework
ENRII	Environment for Natural Resource Information Integration (Qld)
EPA	Environment Protection Agency (Qld)
EPSG	European Petroleum Survey Group (authoritative source of definitions for coordinate reference system information).
EROS	Earth Resources Observation Systems
EU	European Union
FGDC	Federal Geographic Data Committee (USA): see http://www.fgdc.gov/
GA	GeoScience Australia
GIA	Government Information Architecture (DIIE , Qld): see http://www.iie.qld.gov.au/comminfo/gia/
GILS	Government Information Locator System (USA) see http://www.access.gpo.gov/su_docs/gils/
GINIE	Geographic Information in Europe (EC): see http://www.ec-gis.org/ginie/
GIS	Geographic Information System
GML	Geography Mark-up Language (OGC): see http://www.opengis.org/docs/02-023r4.pdf
GNIS	Geographic Name Information Service
GOS-P	Geo-spatial One-Stop Portal (US Government): see http://www.geo-one-stop.gov/
GPATS	Geographical Positioning and Tracking System
GSDI	Global Spatial Data Infrastructure
GSL	Government Services Locator
HDF	Hierarchical Data Format: see http://hdf.ncsa.uiuc.edu/
HTML	Hypertext Mark-up Language (W3C): see http://www.w3.org/MarkUp/
http	Hypertext Transport Protocol (W3C): see http://www.w3.org/Protocols/
ICC	Incident Control Centre
IETF	Internet Engineering Task Force: see http://www.ietf.org/

IPA	Integrated Planning and Assessment (Qld)
IQ	Information Queensland
ISCGM	International Standing Committee for Global Map http://www.iscgm.org/
ISO	International Organization for Standardization http://www.iso.org/
LandXML	An XML schema for land planning: see http://www.landxml.org/
LDAP	Lightweight Directory Access Protocol: see http://www.openldap.org/
LGA	Local Government Authority
MIME	Multipurpose Internet Mail Extension (IETF)
NAP	National Action Plan
N-A	OGC-Australasia Notional Architecture for a Geo-enabled enterprise portal platform
NCSA	National Center for Supercomputing Applications
NDVI	Normalised Difference Vegetation Index
NIST	National Institute of Standards and Technology (USA): see http://www.nist.gov/
NR&M	Department of Natural Resources and Mines: see http://www.nrm.qld.gov.au/
NSDI	National Spatial Data Infrastructure initiatives
OASIS	Organization for the Advancement of Structured Information Standards: see http://www.oasis-open.org/who/
OCS	OpenGIS Catalog Services (OGC)
OECD	Organization for Economic Cooperation and Development
OESR	Office of Economics and Statistical Research (Qld)
OGC	Open GIS Consortium (USA): see http://www.opengis.org
OgcRIM	OGC registry information model
OSI	Open Source Initiative
OSS	Open-source software
OWS	Open Web Service (OGC)
PDA	Personal Digital Assistant
PKI	Public Key Infrastructure (NIST): see http://csrc.nist.gov/pki/
PNG	Portable Network Graphics (W3C) see http://www.w3.org/Graphics/PNG/
QIO	QSIIS Information Office
QPW	Department of Public Works (Qld): see http://www.publicworks.qld.gov.au/
QSIIC	Queensland Spatial Information Infrastructure Committee
QSIIS	Queensland Spatial Information Infrastructure Strategy: see http://www.qsiis.qld.gov.au/
RECC	Regional Emergency Co-ordination Centre

RFP	Request for Participation
RIS	Regional Information Service (Qld)
ROSI	Roster of Strategic Information (OESR , Qld)
SAML	Security Assertion Mark-up Language (OASIS)
SCOTS	Standards-based Commercial Off The Shelf software
SDI	Spatial Data Infrastructure
SDTS	Spatial Data Transfer Standard (USGS): see http://mcmweb.er.usgs.gov/sdts/
SECC	State Emergency Co-ordination Centre
SFS	Simple Features Specification (OGC): see http://www.opengis.org/techno/specs/99-049.pdf
SII	Services Integration Infrastructure (Qld)
SLA	Service Level Agreement
SLD	Styled Layer Descriptors (OGC): see http://ogc.cast.uark.edu/tnm/web/sld.html
SMIS	SmartMap Information System (Qld.)
SMS	Style Management Service (OGC)
SOAP	Simple Object Access Protocol (W3C): see http://www.w3.org/TR/2000/NOTE-SOAP-20000508/
SSL	Secure Sockets Layer
SSQ	Smart Services Queensland
TCP/IP	Transport Control Protocol/ Internet Protocol (IETF)
UDDI	Universal Discovery Description and Integration (OASIS): see http://www.uddi.org/
UNEP	United Nations Environment Programme (Kenya): see http://www.unep.org/
UNGIWG	UN Geographic Information Working Group (USA): see http://www.ungiwg.org/
URI	Uniform Resource Indicator (W3C)
URL	Uniform Resource Locator (W3C)
USGS	US Geological Service (USA): see http://www.usgs.gov
W3C	World-Wide Web Consortium: see http://www.w3.org
WA DLI	Western Australian Department of Land Information
WALIS	Western Australian Land Information System: see http://www.walis.wa.gov.au/
WCS	Web Coverage Service (OGC)
WFS	Web Feature Service (OGC)
WFS-T	Transactional Web Feature Service (OGC)
WMC	Web Map Context document (OGC)
WMS	Web Mapping Service (OGC)

WOG-GIS	Whole-of-Government GIS (Qld)
WP	Western Power
WPOS	OpenGIS Web Pricing & Ordering Service (OGC)
WSDL	Web Services Description Language (W3C): see http://www.w3.org/TR/wsdl
XML	Extensible Mark-up Language (W3C): see http://www.w3.org/XML/

2.2 Terminology

Application	A program that performs a specific function directly for a user. Applications can make use of SDI.
Application Profile	A configuration of one or more general standards designed to solve a more specialised problem. An application profile may choose a limited subset of a general standard, or a specific combination of (for example) content standards and implementation platforms. Important types of application profile include: GML Application Language or Schema – an XML language based on GML that defines feature types and other information objects required for an application domain; Catalog profile – a metadata scheme, typically based on a general vocabulary like ISO 19115, which select which components are to be used, and specifies domains for the values of specific properties.
Architecture	The organizational structure and operating environment of a software system, including the relationships between its parts, and the principles and guidelines governing their design and evolution over time.
Binding	A connection between two complementary components, such as: a client and service; data and portrayal rules;
Capabilities schema	XML schema that prescribes and constrains the syntax and vocabulary for the expression of service capabilities in XML.
Capabilities XML	Specific instance of service-level metadata describing a service instance .
Client	A software component or an application that accesses a service. Clients may be categorized in three ways: Thin clients where the client supports only human-interface code, such as a web browser, a text-only interface or a minimal PDA or WiFi handset, and must support non-proprietary standards. Typically, lack long-term memory such as disk drives. Application code and data access both run remotely, and so are entirely dependant on external network connection. Thick clients where the client supports all the human interface and application code, may support some or all data access code, and may support long-term data memory. Human interface code may be entirely customized and not conform to non-proprietary standards. May not even support human interfaces i.e. may be entirely automated emote processes. May operate at times without network connection. Chubby clients have capabilities somewhere on the spectrum between thick and thin clients i.e. may support some application and data code, and may store limited amounts of data. Will usually but not necessarily support human interfaces. May operate well for limited time without network connection.
Component	Software that packages the client or server implementation of a service and can provide the realization of a set of interfaces . A component consists of software code (source, binary or executable) or other equivalents such as scripts or command files.

Conceptual Architecture	An overview of the services , data, technology and institutional environment of SDI. It describes, in general terms, both what the SDI will include, and how it will operate.
Core Service	A service that is critical to the ongoing operations of the SDI. The Service Registry Service is an example of a core service. It is used to keep track of where SDI services can be found, so if this service were unavailable, client applications would not be able to locate the services required in order to complete a task.
Coverage	A continuous representation of a portion of the earth's surface. A coverage may be a collection of features (like a vector dataset) or it may be a raster or gridded surface representing one or more attributes.
Coverage Data	Gridded data, imagery, swath data, and continuous functions. Contains values for every point in a geospatial data. Relevant specifications for Information Queensland coverage data formats may include: • SDTS - Spatial Data Transfer Standard • GeoTIFF - Georeferenced Tagged-Image File Format • HDF – Hierarchical Data Format
Custodian	The authoritative manager of an SDI resource, whether data set, service or component , who is responsible for declaration of the policies regarding utilization and accounting for the resource.
Datastore	Any type of persistent storage for Information Queensland components and data. Content may be static or dynamic. May include database systems, file systems, structured text storage, XML repositories etc.
Event	An occurrence of interest to users or developers of the SDI. Events can be things such as the adjustment of a feature in a framework data layer, a flood in the Red River basin, or the release of a new specification for a SDI service .
Framework Data	The set of geospatial data that provides the reference framework for all other SDI compliant geodata . See the SDI Framework Data Definition . Seven geospatial data themes may be considered to be of fundamental importance to many applications. Known as "Framework Data," these themes are: Elevation, Orthoimagery, Hydrography, Transportation, Government Units (administrative boundaries), Cadastral (property boundaries), and Geodetic Control. Framework Data shall, at minimum, be able to exchange data in a manner that complies with these emerging standards. Information Queensland shall be able to access both Framework Data sources and other, non-Framework data.
Geodata	Georeferenced spatial data such as a road network or a satellite image. Geodata explicitly describes the spatial extent of a set of features or describes a measurable surface. It includes both geospatial data and geolinked data .
Geodata Resource Registry	Peer registries that contain metadata about geodata resources. These registries identify the agencies that are responsible each attribute dataset, and descriptions of that attribute data. They also contain information on how to access the data.
Geolinked Data	Geodata that are referenced to an identified set of geographic features without including the spatial description of those features. Geolinked data are normally attribute data in tabular data (such as population counts) that refer to a known framework (such as municipalities), where their unique identifier (such as the municipality name) refers to the elements (the municipalities). The term “geolinked data” refers to all attribute data that is not directly attached and bundled with the geographic coordinates to which it applies.
Geospatial Data	Geodata with explicit geographic positioning information included, such as a road network from a GIS, or a georeferenced satellite image. Geospatial data may include attribute data that describes the features found in the dataset.
GeoTIFF	Georeferenced Tagged-Image File Format

Information Queensland	The concept of an all-of-government, distributed system of services providing added-value spatial information products to users in all sectors of society in Queensland and beyond
Interface	The named set of related operations that structures how a client interacts with a server . The state and functionality of the underlying component is hidden, and is only made externally accessible through its interface (i.e. the interface is the only "public" or "visible" part of the component). The same interface may be provided by several components and used by many components or applications .
LandXML	Facilitates the exchange of data created during the Land Planning, Civil Engineering and Land Survey process. LandXML is an XML application that utilizes XML Schema to define relationships and custom data types. Much like GML, LandXML is a fairly robust application of XML as a data format, targeted specifically towards land planning, engineering and survey applications.
Library	A type of registry intended for recording references to entity types (as distinct from recording instances) i.e. a look-up. Content is generally static once instantiated
Map	A pictorial representation or portrayal of geographic data
Master System	A primary system that is mirrored to other identical systems , so as to ensure that any important information or services can be accessed in an identical fashion from multiple sites .
Metadata Standard	Data will be documented according the FGDC Content Standard for Digital Geospatial Metadata (CSDGM) and/or the ISO 19115
Notional Architecture	A design framework based on vendor-neutral, open standards.
Operation	An interaction between a client and a server , resulting in a transfer of information or an action. An operation can be either an interrogation (i.e. request-response) or an announcement (i.e. notification).
Portal Services	Provide "one-stop" access to the Provider Services. An online access point to a collection of geospatial data -- more precisely, to a collection of services that provide data or relevant functionality. A Portal does not store or maintain the data and its associated services; rather, these are distributed in many computers nationwide and maintained by the agency or organization that is responsible for its data and services. A Portal will have a level of functionality necessary to provide access to the provider service. This may involve visualisation and processing, or may merely be links to other interfaces to the provider services. Portals shall be based on open standards and specifications that are defined collaboratively by a various interested parties, are freely published, and are able to be implemented by any vendor or organization.
Portrayal Binding	Binding of a WFS or WMS with style information to provide standard portrayal. See WMC.
Primary System	A system that provides a core service and thus is critical to the operations of the SDI. Although a primary system may not be mirrored to other sites , or even the most popular implementation of a particular service , they are publicly identified and funded to ensure that core services are implemented on at least one system .
Process	A system offering a single interface for the execution of a single task creating one or more products. Processes may be grouped to provide a service.
Provider Services	Include map, data, gazetteer, catalogue and any other services hosted and maintained at remote sites independently of Information Queensland and are accessed by the it. The role of the Portal Services is to provide "one-stop" access to the Provider Services.
Reference Architecture	A technical blueprint that identifies and defines the services that comprise the SDI, and specifies the interfaces to those services .

Registry	A listing of the specific, individual services, components, datasets, or other things that comprise the SDI or are relevant to its users. Instance registries are used to identify, locate, and describe individual instances. Many registries refer to associated Type Libraries that record the allowed types within registry classes e.g. types of services, types of user authorities.
Request	Invocation of a server operation by a client
Response	Result of an operation returned from a server to a client
Schema	A schema is an expression of the type using a particular data modeling language. Types provide the basis for classification taxonomies for sets of schema definitions. In the OGC the application data-modeling language is GML and each schema fragment corresponding to a given type is defined in GML.
Server	(a) A software component that delivers a service. (b) A physical implementation of such a component, that provides the realization of its operations.
Service	A collection of operations, accessible through one or more interfaces, that allows a user to evoke a behaviour of value to that user. A server is delivers each service. A service may encapsulate may processes. A "service instance" is another name for a server (b).
Service binding	Specific syntax and parameter values used by a client to invoke a specific server operation
Service capabilities	Service-level metadata describing the types , operations , content , and bindings available at a service instance . Organization, classification, and presentation of those entities may also be conveyed by the capabilities information.
Service Registry	Contains metadata about individual services, including the type of service, and information about how to access the service. A Service Registry Service provides a mechanism to: 1. Register the characteristics of certain types of services (in a Service Type Registry) and register the existence of specific services (in a Service Registry); 2. Search Service Registries; and 3. Retrieve detailed information (metadata) that describes a service, in order to evaluate its suitability to meet a need. A Service Type Registry contains information about specific <i>types</i> of services, including the name of the service type, and its definition. A Service Type Registry is being maintained manually, at http://www.geoconnections.org/architecture/systems/ServiceTypeRegistry
Spatial reference system (SRS or CRS)	A projected or geographic coordinate reference system
Styled Layer Descriptor	Metadata describing presentation of geographical types for portrayal in WMS and WFS Addresses the need for geospatial consumers (either humans or machines) to control the visual portrayal of data. It can be used to portray the output of Web Map Servers, Web Feature Servers and Web Coverage Servers. Uses consistent XML-based methods persisting in an SDI library.
System	Servers and data organized to accomplish one or more Services . A system may be accessible at more than one site .
Type	A classifier for a kind or class of geographic entity. Types are meaningful elements in a particular universe of discourse.

Web Agent	People or software that act on the web information space. Software agents include browsers, servers, proxies, spiders, and multimedia players mediating interactions on behalf of a person, entity, or process.
Web Service	Application logic accessible across a network using standard Internet protocols. Web Services combine the best aspects of component-based development and the Web. Like components, Web Services represent functionality that can be easily reused without knowing how the service is implemented. Unlike current component technologies that are accessed via proprietary protocols, Web Services are accessed via ubiquitous Web protocols (e.g. http) using universally accepted data formats (e.g. XML).

3 Introduction

3.1 *Purpose of this document*

This document provides a report to ASIBA on technical architecture for the SIDP project. The primary goal is to

- Describe components and the connections required for a distributed service-oriented architecture based on relevant open standards
- Outline event traces that map the SIDP scenarios onto the technical architecture
- Specify necessary profiles, vocabularies, and specialized encodings required for communication on the interfaces

An additional goal is to identify components that may be developed and deployed during the SIDP project, but may be left behind after the project. These may contribute core components to the development of an Australian Spatial Data Infrastructure.

The present document draws on and should be read in conjunction with the following:

1. the OGC-A Notional Architecture for a Geo-enabled enterprise portal platform [14]
2. scenarios, requirements, content and gap analysis provided by SIDP [17]

It can be regarded as one realization of the Notional Architecture but is not necessarily closely mapped.

The document primarily serves to enable writing RFP (Requests for Participation) and development of a demonstrator project.

3.2 *Relationship to OGC-A Notional Architecture*

The OGC-A Notional Architecture (N-A) [14] describes a service architecture for geospatial information delivery and processing. The functions in the architecture are associated with a fine decomposition of service components. The use of standard interfaces allows these to be coupled in a variety of configurations as needed to create complex applications. The prototype application is an information portal.

The SIDP is complementary to this approach, being focussed around specific scenarios. These involve rich interactions, and significant performance and reliability requirements. Development of an architecture for SIDP will lead to the examination of configurations designed to service these scenarios, and requires increased focus on the client and operator. SIDP also has a goal of primarily demonstrating the capabilities of existing off-the-shelf products. Software development is to be primarily confined to component configuration. In some cases this has led to simplifying or modifying specific interactions within the scenarios to better match the expected current software capabilities.

4 General implementation patterns

4.1 Application Boundary Concept

An API (Application Programming Interface) is a view from the outside at some component. We have an *Application Boundary* which is the place at which *services* are delivered. Whilst the Notional Architecture [14] refers to the services being used by “standard web clients”, for a system to illustrate the scenarios the architecture needs to specify more detail of how the demonstrators should be built. This requires consideration of the application realm.

A summary architecture for SIDP is shown in Figure 1. This shows two types of application boundary:

- 1 Standard interfaces on which the client application accesses services through OGC Web Service interfaces (see Table 1, Table 2)
- 2 Interfaces for which there is currently no standard OGC specification

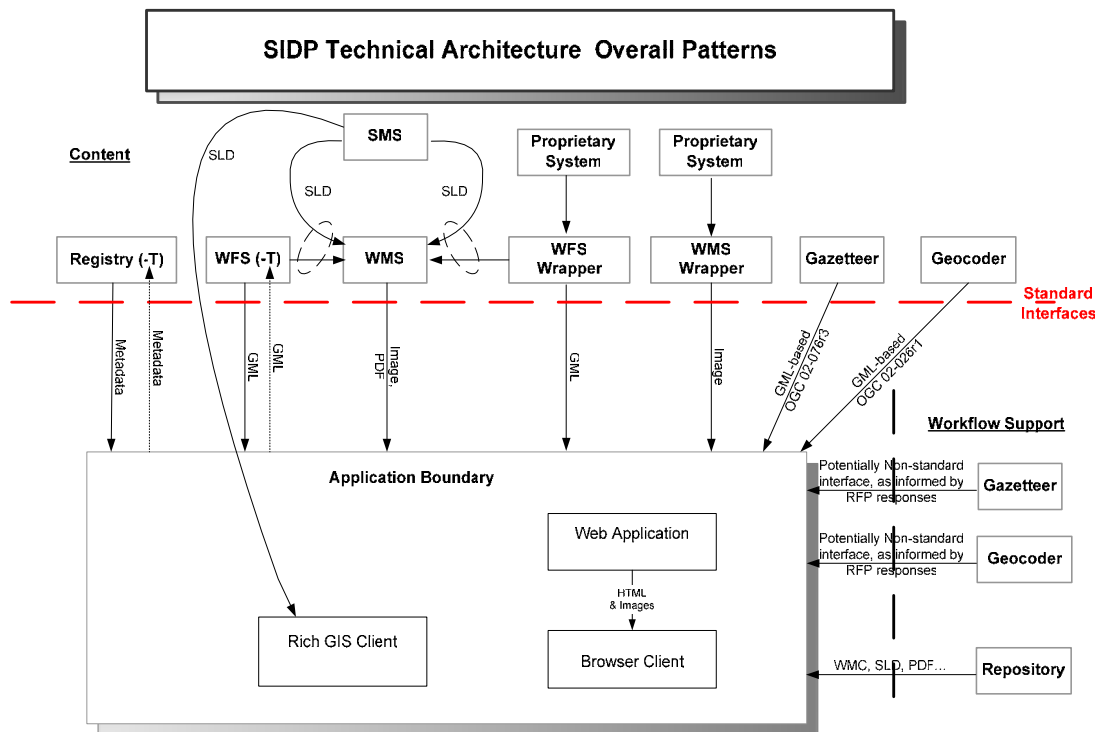


Figure 1 Service Overview and Application Boundary

Figure 1 is a generalisation of specific portal architectures, such as the GOS-P architecture shown in Figure V2-4 in the OGC-A Notional Architecture [14]. In this version of the diagram we have collapsed various client roles into a generic “rich client”.

Note that the provision of web-access to information services should not be intrusive on the normal data management procedures of a custodian organisation. The web-service interfaces should be seen as value-adding, enabling use of information resources by a wider group of users, and their integration into systems that cross organisational boundaries, and are accessed through a flexible set of client applications. Certain functions, particularly in content management and maintenance, would be expected to require interfaces with high degrees of functionality that are only likely to be available through tightly coupled, proprietary applications for the foreseeable future. Other considerations, such as specific workflow, security, and dependencies on other data capture and management systems may also indicate the use of private interfaces. These would normally be contained within organisational boundaries, so will not impinge on scenarios such as the ones considered for SIDP.

4.2 Web service interfaces

Services described will be implemented as wrappers or façades to storage and processing applications, exposing them through standard http-hosted interfaces. It is not necessary for the client to have knowledge of implementation details behind the façade. All required operations are enabled through the interface.

The interfaces are defined in terms of standardized messages transported over http. Communication is normally via http GET and POST, though SOAP wrappers may be used if both client and server support them.

The use of http provides that each service instance is listening for requests at a specified URL address. Following standard web terminology, the component that initiates an operation is designated the “client” while the component that responds is the “server”. These roles strictly apply only to a particular transaction. A single application may play different roles in different transactions, particularly involving different partner components. For example, in order to present content integrated from more primitive services, a “cascading web map server” acts as a client to upstream map services, requesting content from them and then serving this to the ultimate client.

The use of a message-based interaction style (sometimes known as Representational State Transfer or REST), rather than a Remote Procedure Call (RPC) has an advantage that a *static document*, with the same format as the service response, can be used in testing and in preliminary deployment. For example, for some of the services indicated in the implementation architecture here, in the short term there may only be a single resource available. It may be convenient, therefore, to initially implement the architecture by storing this resource as a static document and serving this from a file-system or a simple web-page server, rather than through a complete web-service application. Replacing a dynamic service by a static file with the format of the dynamic service response is known as “stubbing” the service. By implementing the client application to use the standard document format, the architecture is pre-enabled for when dynamic services are deployed.

This approach is consistent with the standard web-architecture.

4.3 OGC Web Services

The implementation architecture is constructed from components that are coupled on **standard interfaces**, as per the *OGC-A Notional Architecture* [14]. In particular, whenever transactions occur across organisational boundaries then use of standard interfaces insulates the parties from dependencies on specific software strategies being used by their partners.

Furthermore, even when the transactions occur *within* an organisation, leading to a situation where a proprietary interface might be considered, **standard interfaces should be preferred whenever feasible**. This allows easier replacement or addition of service components as the implementation evolves, where the new services may be either internal or external to the organisational boundary.

As far as possible the interfaces specified for implementation are drawn from the set of *Adopted Implementation Specifications* issued by Open Geospatial Consortium, with some associated standards and specifications from ISO/TC 211. To satisfy the required functionality in a few areas, some less mature specifications are indicated. Collectively, these are referred to as “OGC Web Services” or *OWS* [15]. Applications that implement the standard interfaces are termed “standards-based software”. Note that this constitutes a small subset of the components specified in the OGC-A Notional Architecture, being that subset for which it is reasonable to expect standards-based commercial off-the-shelf (SCOTS) implementations to be available.

For WMS and WFS interfaces, software compliance tests are available from the OGC CITE website (<http://cite.occamlab.com/>).

In the service overview (Figure 1) some WMS and WFS services are shown as “standalone” components, while another set of services are labelled as **wrappers**. By implication the standalone services are where the OWS interfaces are closely coupled to a private datasource. The “wrappers” are intended to suggest that some public but non-standard datasources might be amenable to “wrapping” to give them a OWS interface, where the wrapper is not tightly coupled, and might be provided from a remote site. This strategy has been employed in a number of cases in OGC IP projects where a datasource was exposed through a non-standard but well-behaved web-page, and a “screen-scraping” approach used to transform the information from HTML into a standard XML form. **Implementing WMS or WFS wrappers might be an opportunity for an SIDP participant to provide a value-adding service.**

Within the scenarios there are places where interactions are required for which there is no standard OWS interface, so a proprietary interface must be used. We have attempted to ensure that communication on proprietary interfaces is constrained to occur only within an organisation.

4.4 Standard interfaces for SIDP

Table 1 lists the OWS interfaces to be used in the SIDP implementation, mapped to functions, and with links to the relevant section of the OGC-A Notional Architecture. In particular, the entries in the table specify variants and versions of the interfaces required to satisfy the scenarios. The full citations for the interface specifications are included in the Normative References (§**Error! Reference source not found.**).

Table 1. OWS interfaces and versions required for SIDP architecture

Function	Interface(s), Version	Notional Architecture
Map service	SLD-enabled WMS - WMS v1.1.0 or later [23],[24],[25] - SLD v1.0.0 [18] Cascading SLD-enabled WMS (for feature portrayal) - WMS v1.1.0 or later [23],[24],[25] - SLD v1.0. 0 [18] - WFS v1.0. client [21]	V3-8 Portrayal Services, §8.2 Cascading SLD-enabled Web Map Service
Map viewing	Context-enabled WMS client - WMS v1.1.0 and later [23],[24],[25] - WMC v1.0.0 [22]	Not explicitly identified in Notional Architecture. See general reference to Thick Client in V2-2, §2.3 Architecture Component Overview, and Web Agent in V2-2, §2.5
Style management: Symbol Catalog Symbol Repository Style Catalog Style Repository Style-bindings catalog Style-bindings Repository	CS-W - CS v2.0 [3] - SLD v1.0.0 [18] (for WMS & WFS) <i>or</i> XSLT+SVG (for WFS only)	V3-5 Data Repository Service Components, §5.4.8 Map Style Library, §5.4.9 Map Symbol Library, §5.5.8 Map Style Registry, §5.5.9 Map Symbol Registry, V3-9 Registry and Catalogue Service Components §9.14 Map Styles Library Service, §9.15 Map Styles Management Service, §9.16 Map Styles Registry Service

Context management: Context Catalog Context Repository	CS-W - CS v2.0 [3] - WMC v1.0.0 [22]	V3-5 Data Repository Service Components, V3-9 Registry and Catalogue Service Components
Feature data access service	WFS - WFS v1.0.0 [21] - supporting externally defined GML Application Schema - GML v2.1.2 or GML v3.1 [9],[10],[11] Filter - Filter Encoding v1.0.0 [6]	V3-6 Data Services Components, §6.8 Web Feature Service. This will be supported by V3-5 Data Repository Services Components, §5.3.3 Feature Datastore
Feature data upload service	WFS with Transactions - WFS-T v1.0.0 [21] - supporting externally defined GML Application Schema [9],[10],[11]	V3-6 Data Services Components, §6.8 Web Feature Service. This will be supported by V3-5 Data Repository Services Components, §5.3.3 Feature Datastore
Feature data request	WFS client - WFS v1.0.0 [21] - SLD v1.0.0 [18] for portrayal - Filter Encoding v1.0.0 [6]	Not explicitly identified in Notional Architecture. See general reference to Thick Client in V2-2, §2.3 Architecture Component Overview, and Web Agent in V2-2, §2.5
Feature data update request	WFS with Transactions client WFS-T v1.0.0 [21]	Not explicitly identified in Notional Architecture. See general reference to Thick Client in V2-2, §2.3 Architecture Component Overview, and Web Agent in V2-2, §2.5

Discovery service & client – datasets, services, other resources	CS-W - CS v2.0 [3] Filter - Filter Encoding v1.0.0 [6]	V3-5 §5.5 Registry Services; V3-9 Registry and Catalogue Service Components, in particular V3-9 §9.9 “Geodata Resource Registry Service
Publishing service & client – datasets, services, other resources	CS-W with Transactions CS v2.0 [3]	V3-5 §5.5 Registry Services; V3-9 Registry and Catalogue Service Components, in particular V3-9 §9.9 “Geodata Resource Registry Service

4.4.1 Notes:

1. The N-A proposes a single service interface for map portrayal – the WMS. The SIDP event traces generally follow this recommendation, though it is also recognised that
 - a. portrayal of feature-data requires that the WMS itself act as a client to WFS services that are the source of the feature data
 - b. a rich client may access dataservices directly and supply its own portrayal service
2. The WMS may invoke other services to obtain a particular portrayal binding. The Style Management system is a subsidiary architecture, that can be composed by choreographing repositories, registries, and middleware indicated in the N-A. Two OGC Discussion Papers on “Style Management Services” (see **§Error! Reference source not found.**) present a preliminary system design. There is additional discussion in §7.5
3. The portrayal services (WMS, rich WFS client) shall be responsible for required **coordinate transformations**.
4. CS-W is indicated for several catalogue, and also repository, services. The rationale for this, and implementation alternatives, are discussed in §7.1.

4.5 Additional interfaces for SIDP

There are also requirements for a number of services which for which there is no Adopted Implementation Specification currently available, but whose pattern closely resembles existing services. In these cases the missing service can sometimes be implemented as a “profile” or adaptation of a more general service.

Table 2. Provisional service interfaces – proposed but not mandated for SIDP

Function	Potential interface, Version	Notional Architecture
Gazetteer service & client	OGC Gaz profile of WFS (v0.0.9) [7]	V3-7 Geodata Processing Service Components, §7.10 Gazetteer Service, supported by V3-5 Data Repository Services Components, §5.3.4 Gazetteer Datastore.
Geocoder service & client	OGC GeoC profile of WFS (v0.7.6) [8]	Not explicitly described in OGC-A Notional Architecture.
Document service (e.g. for static map products)	WMS with specialised/limited Capabilities [23],[24],[25]	V3-8 Portrayal Services Components

5 Event Diagrams

5.1 Notation

We present a series of diagrams to elaborate sequences of events involving different services. The intent is generally to separate out significant sequences that are independent. The level of detail corresponds with the decomposition of the scenario flow as interactions between pairs of components.

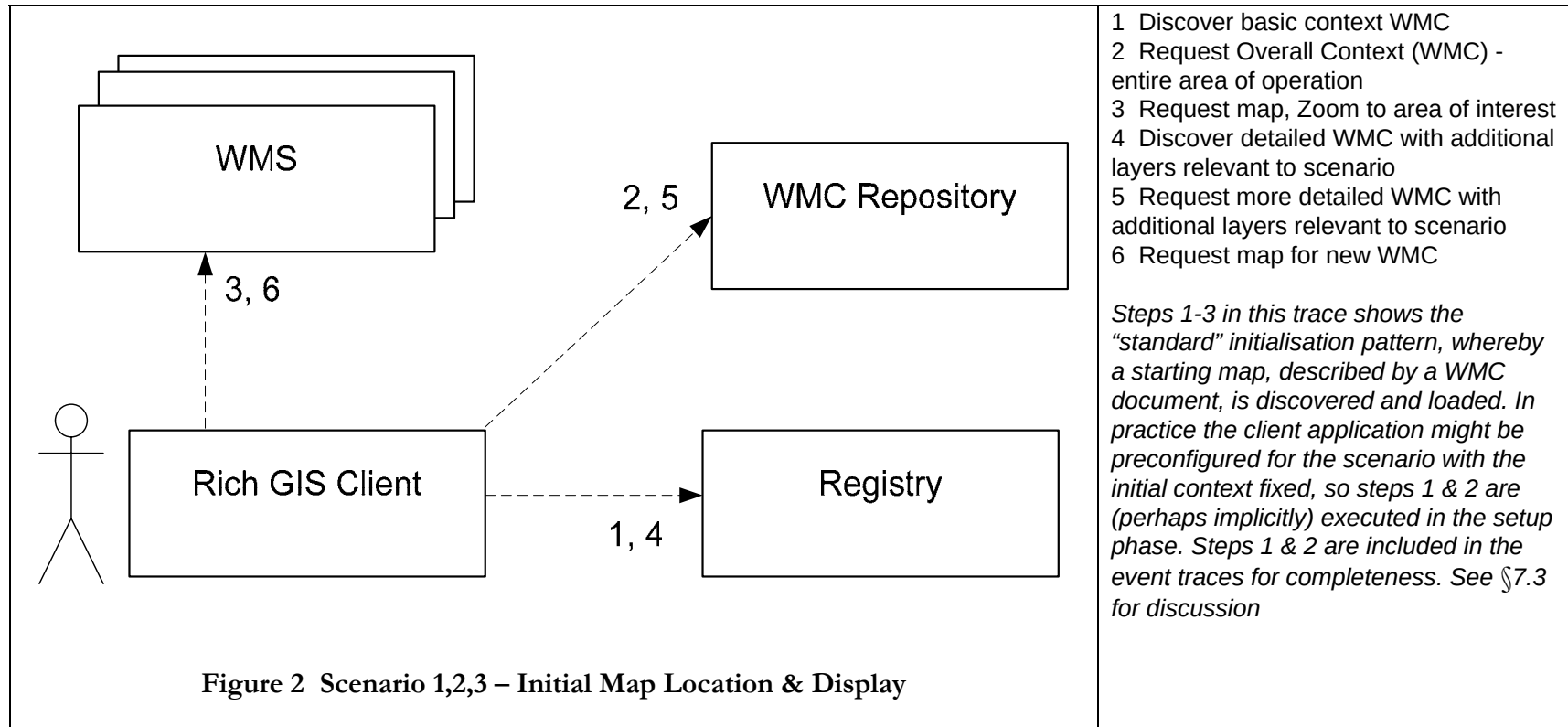
The event sequences are drawn using the BON notation **Dynamic Diagram** style which allows for a 2D layout showing communicating entities and a numbered list of events as a legend. This effectively maps the scenario onto the architecture. These are invoked in a certain order in the narrative, but that order is not technically implied by the architecture and would not always be followed in real use.

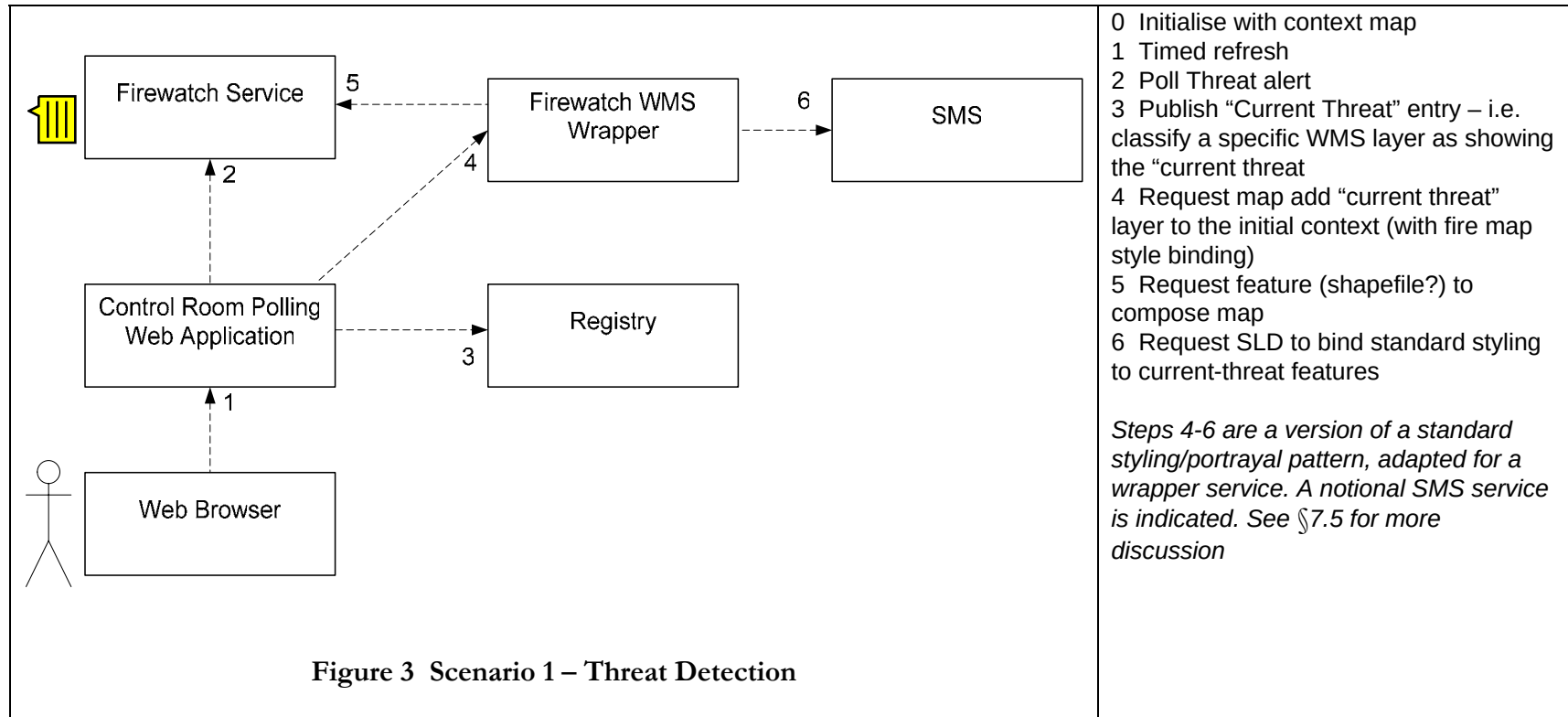
Where possible, the interactions use operations defined as part of standard interface definitions. Even where it might be possible within an organisation or between organisations using applications from the same vendor, the technical architecture does not specify tight coupling of services on proprietary interfaces, unless it is essential either for technical implementation reasons or to support the scenarios.

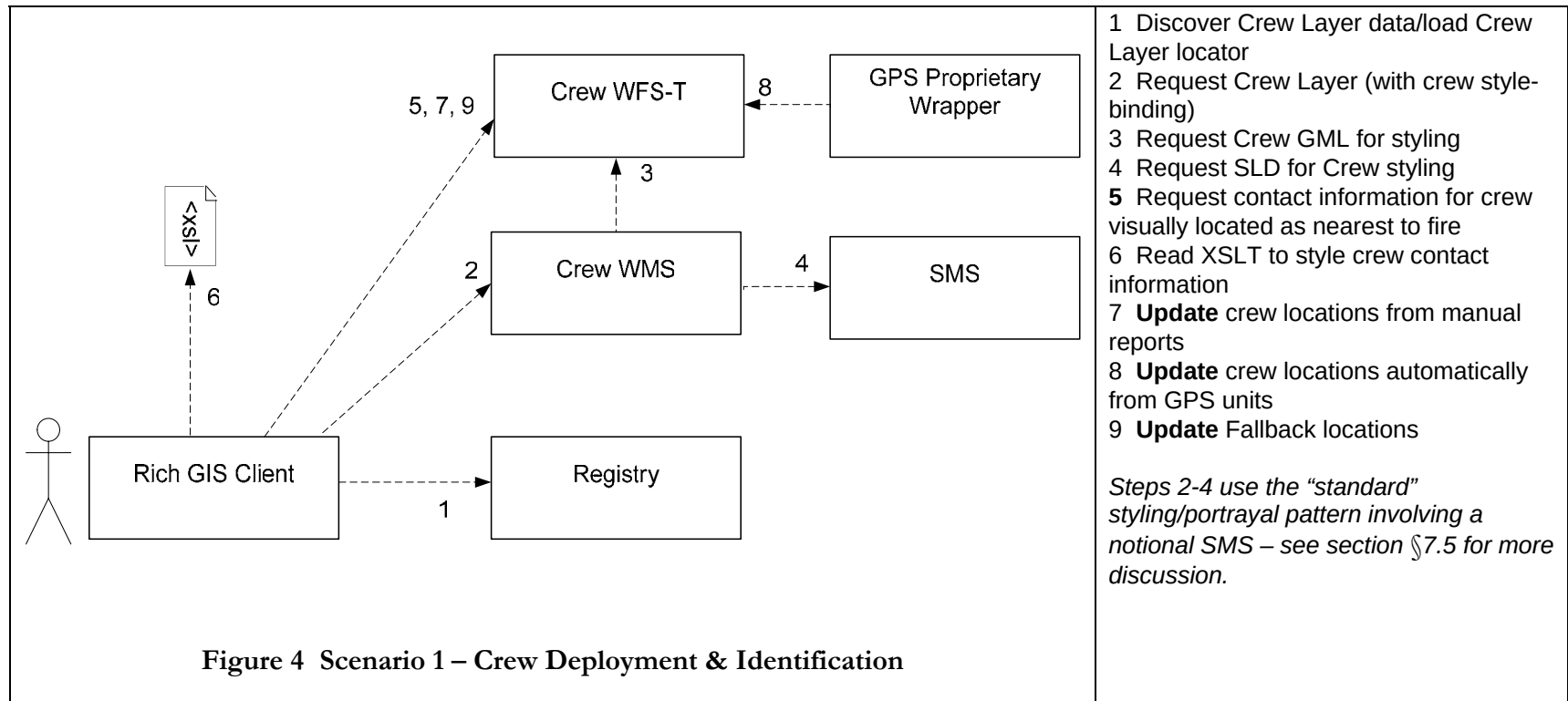
The complete Component Architecture for the SIDP is the **union** of the component diagrams shown in this section. In most cases there is also one or more associated content standards that are indicated in the accompanying text.

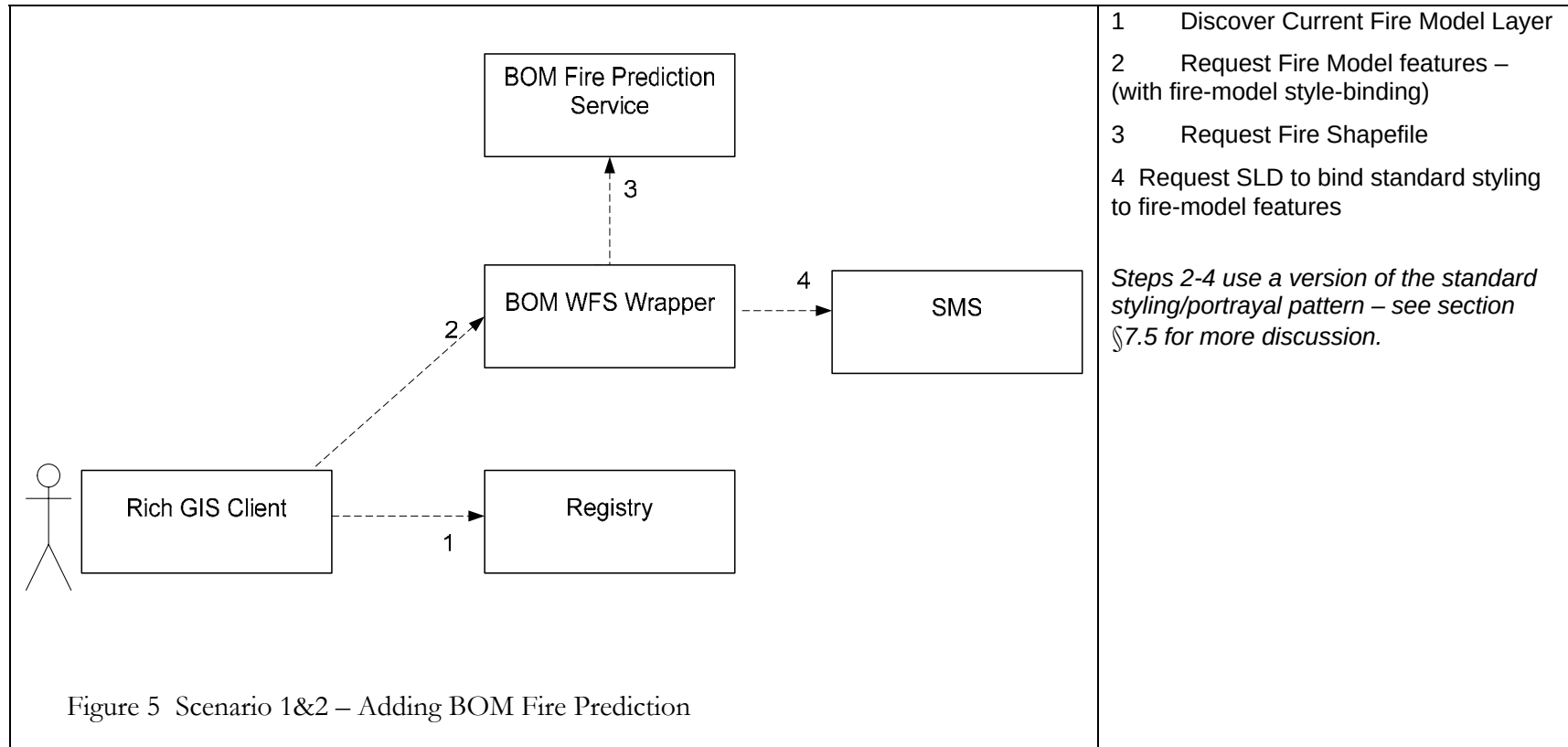
5.2 Granularity

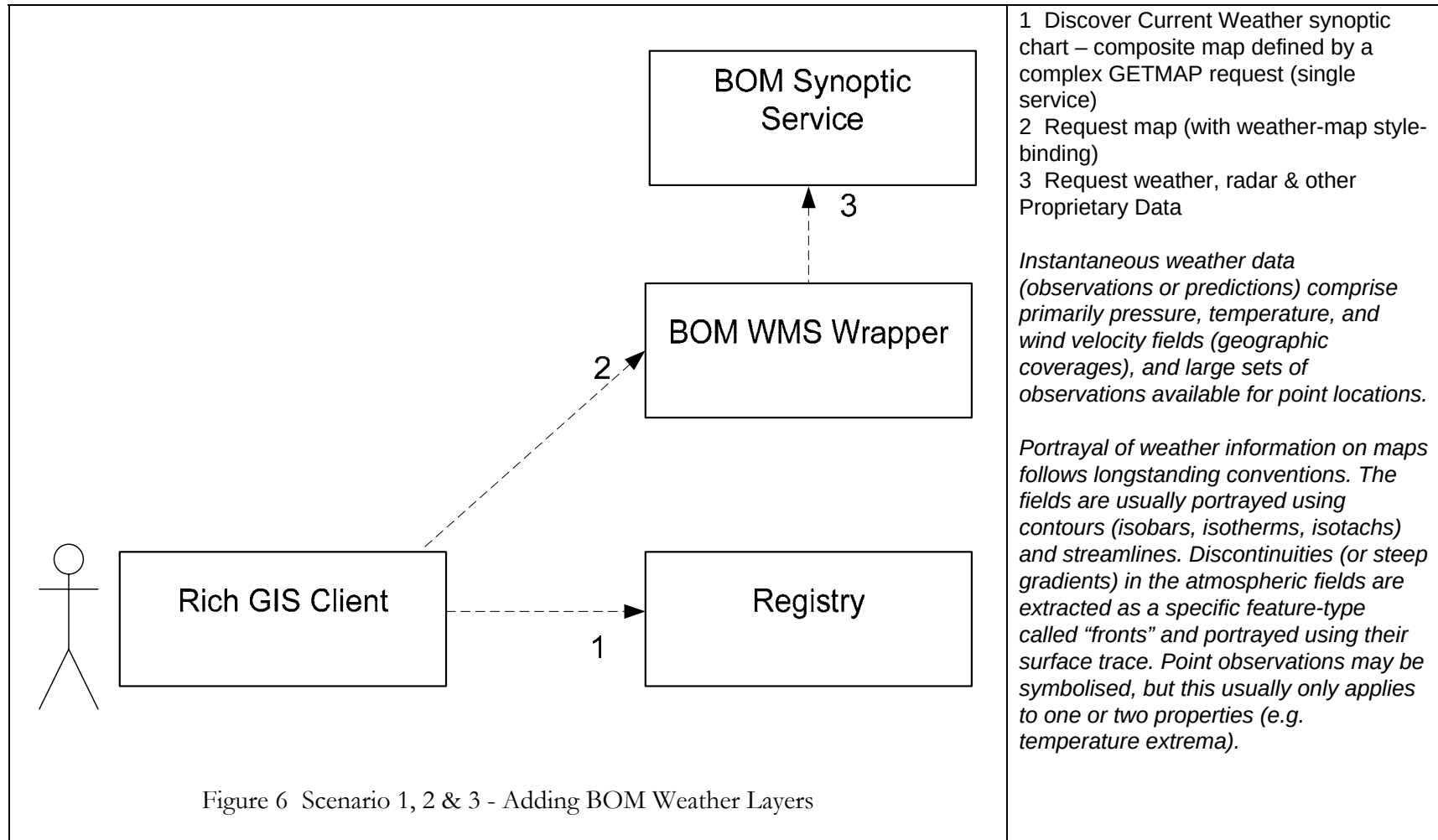
The event diagrams indicate interactions between service components at a level of detail that stops short of the sequence diagrams that will fully describe an engineering deployment. Each of the arrows in a BON diagram implicitly encapsulates a series of messages that might appear in a UML sequence diagram. The exact sequence will depend on details of the capabilities of the components, primarily the client software.

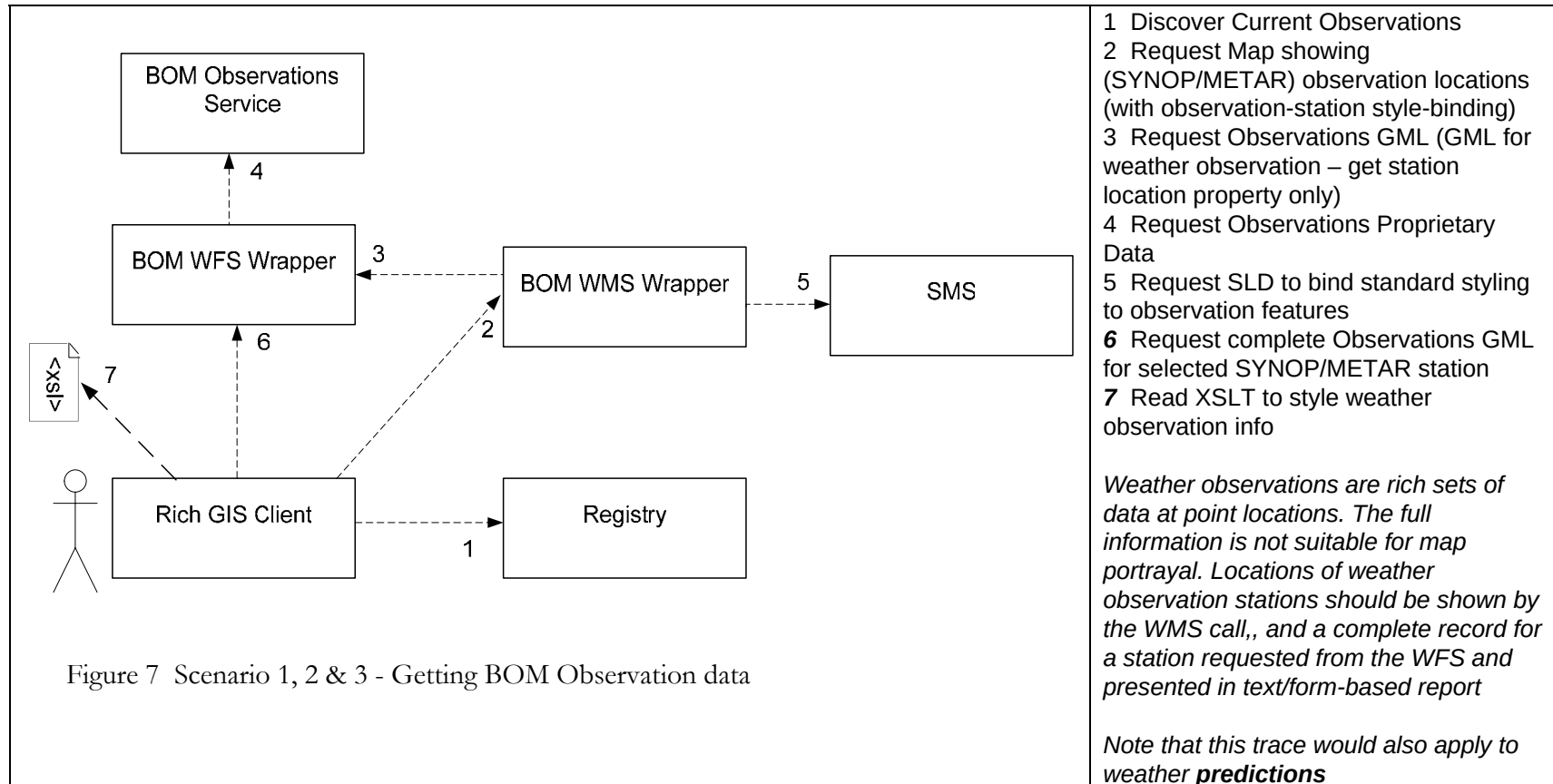


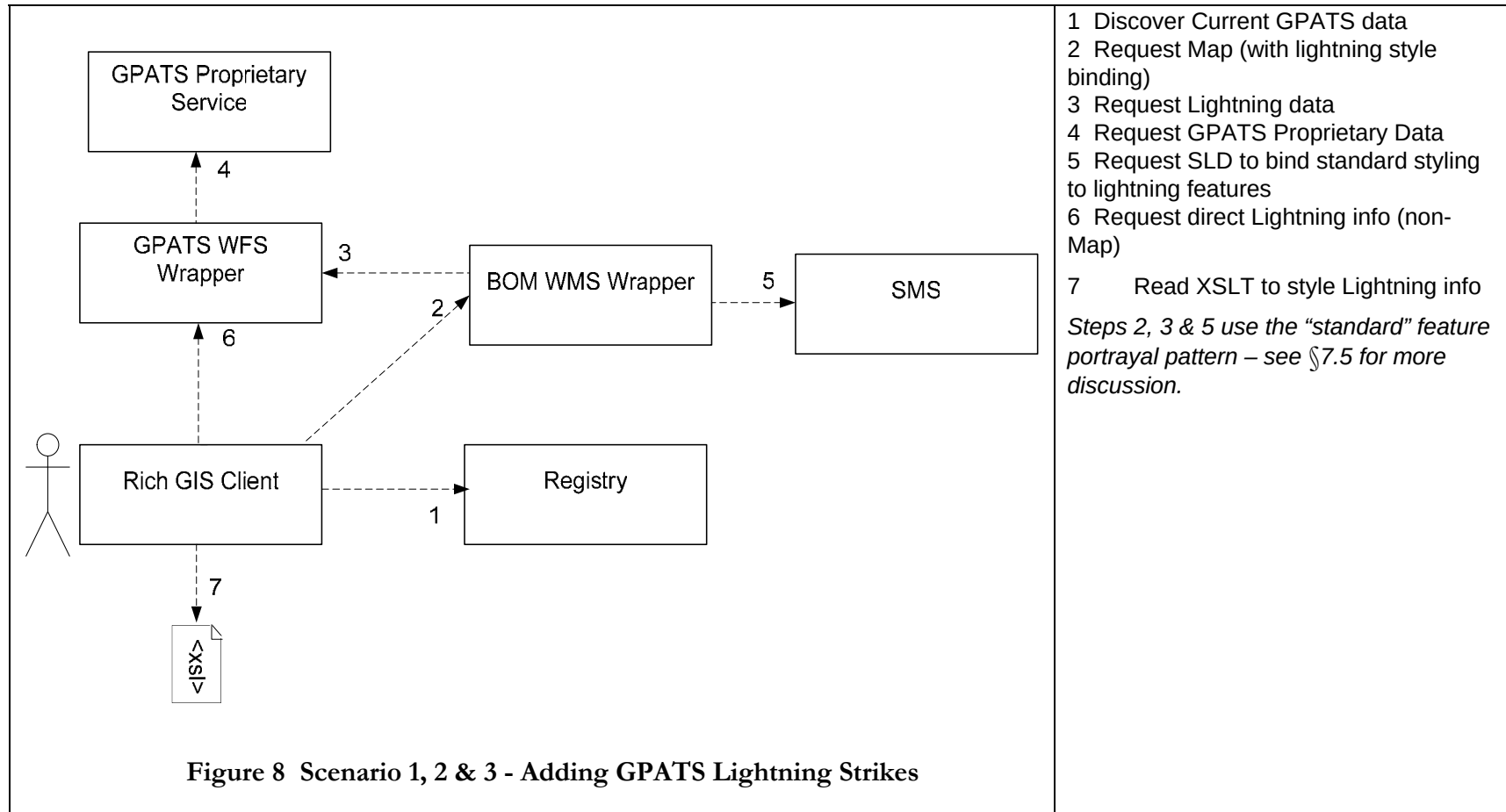


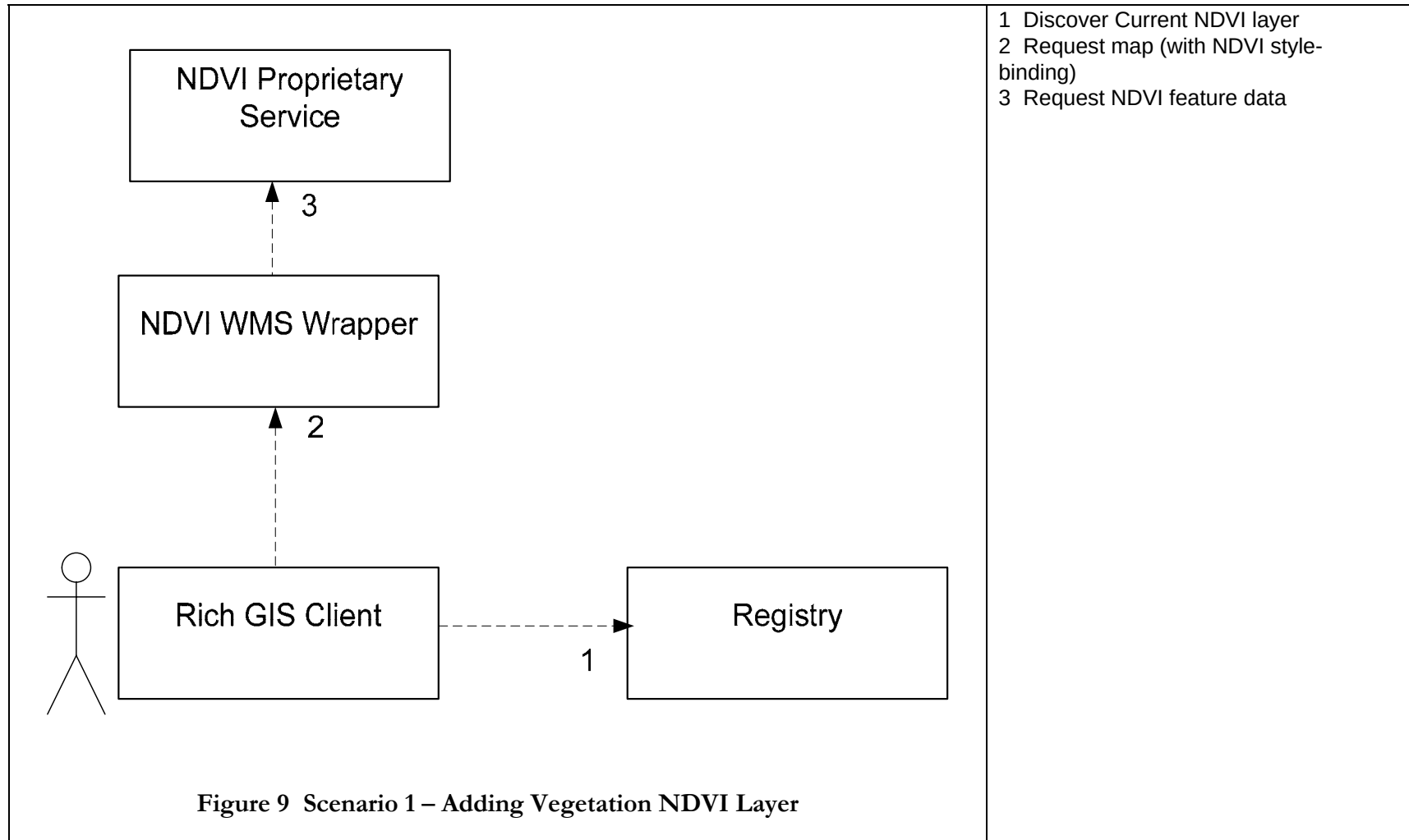


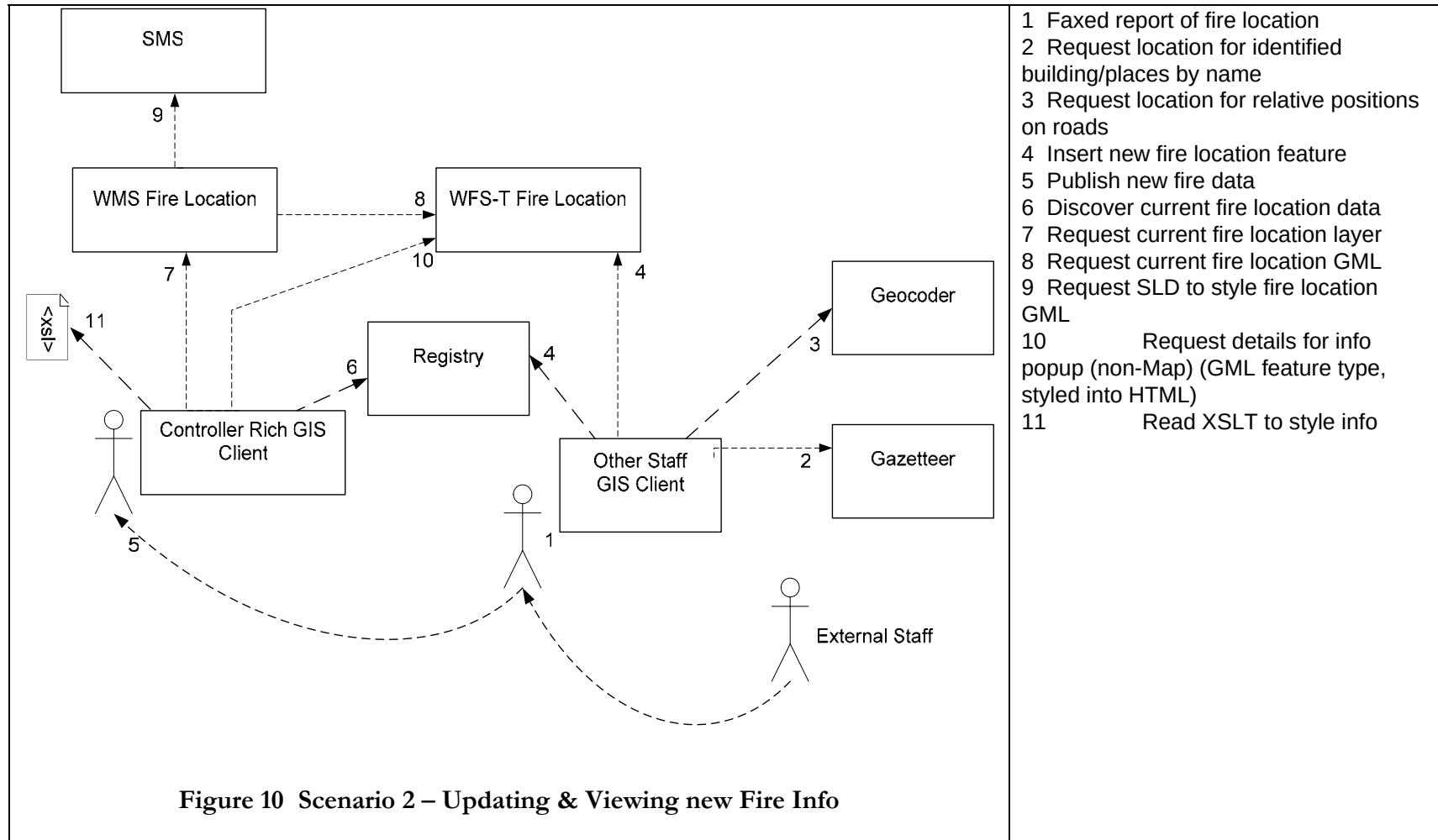


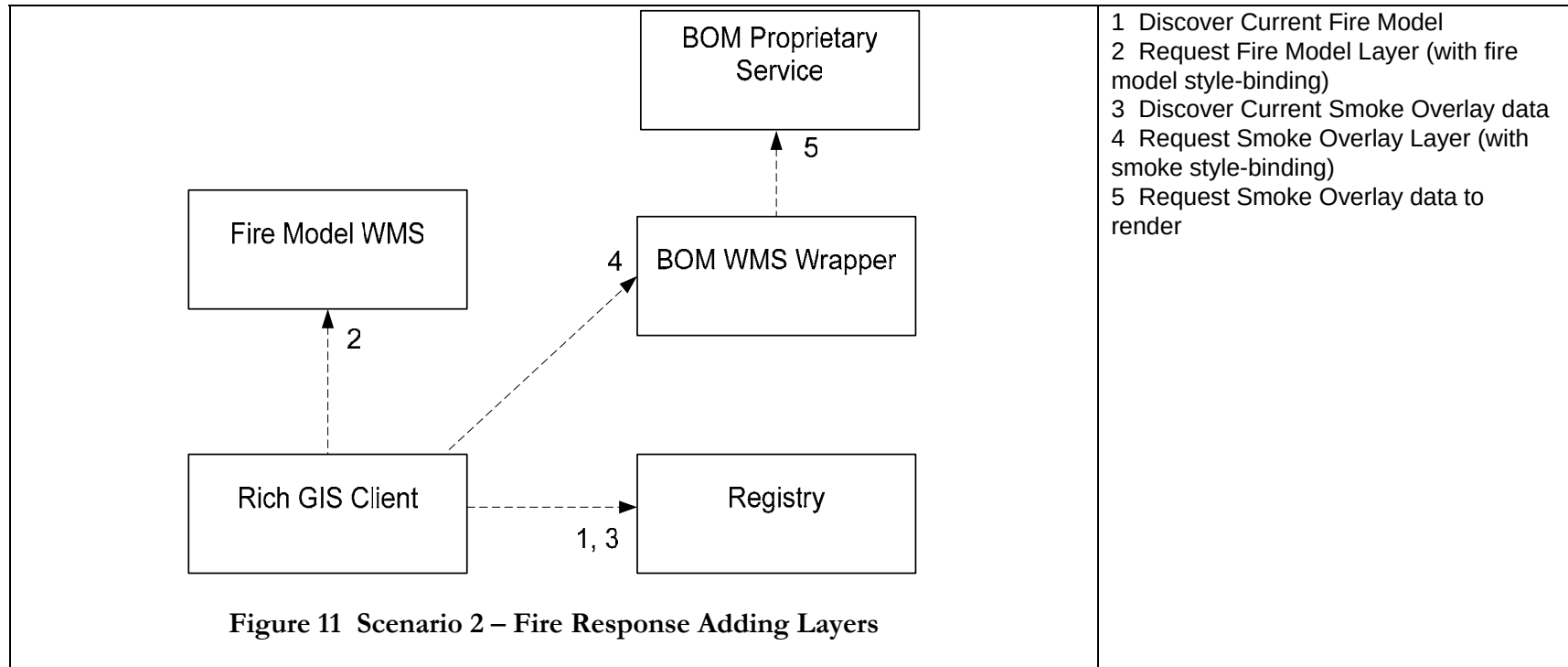












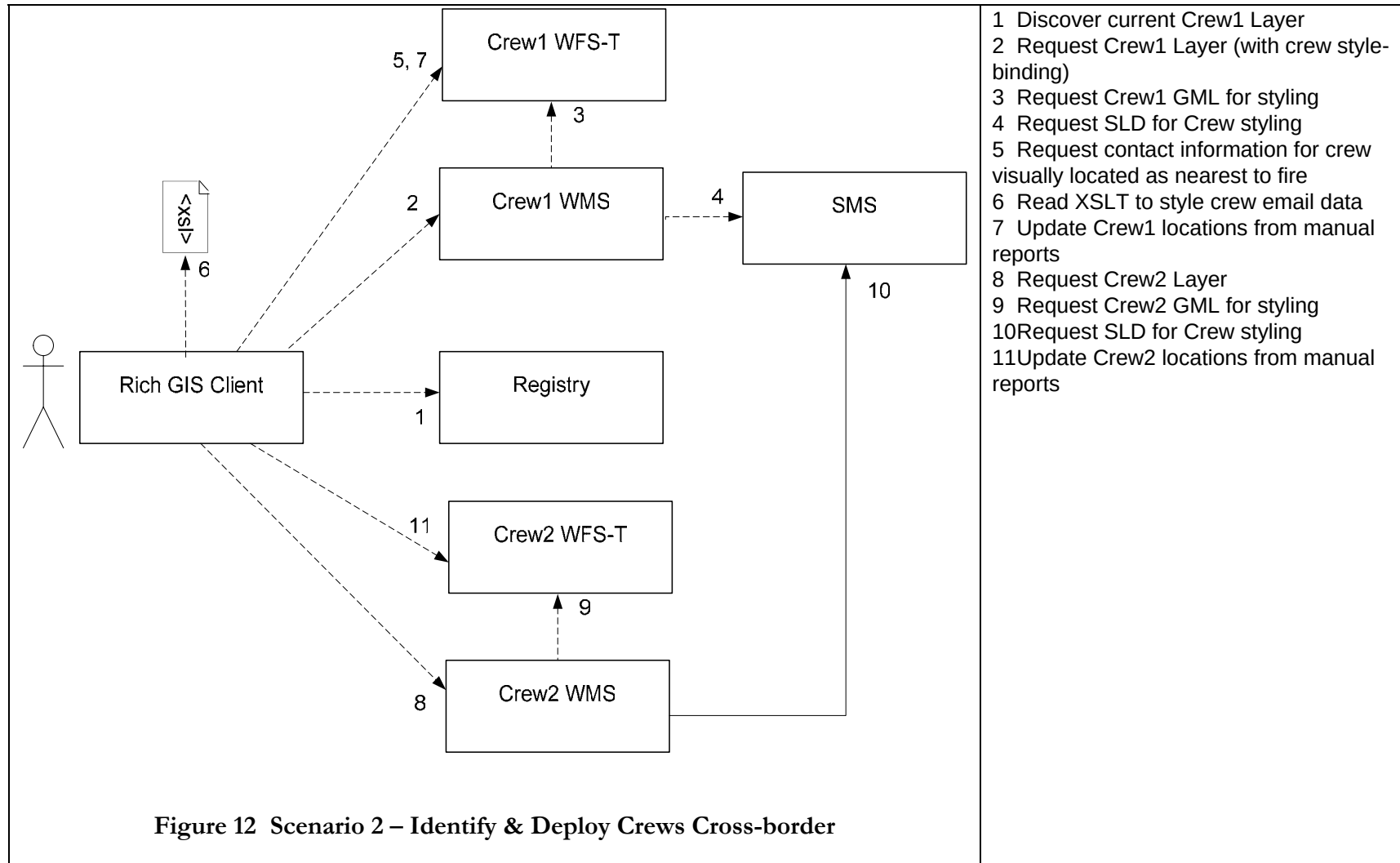
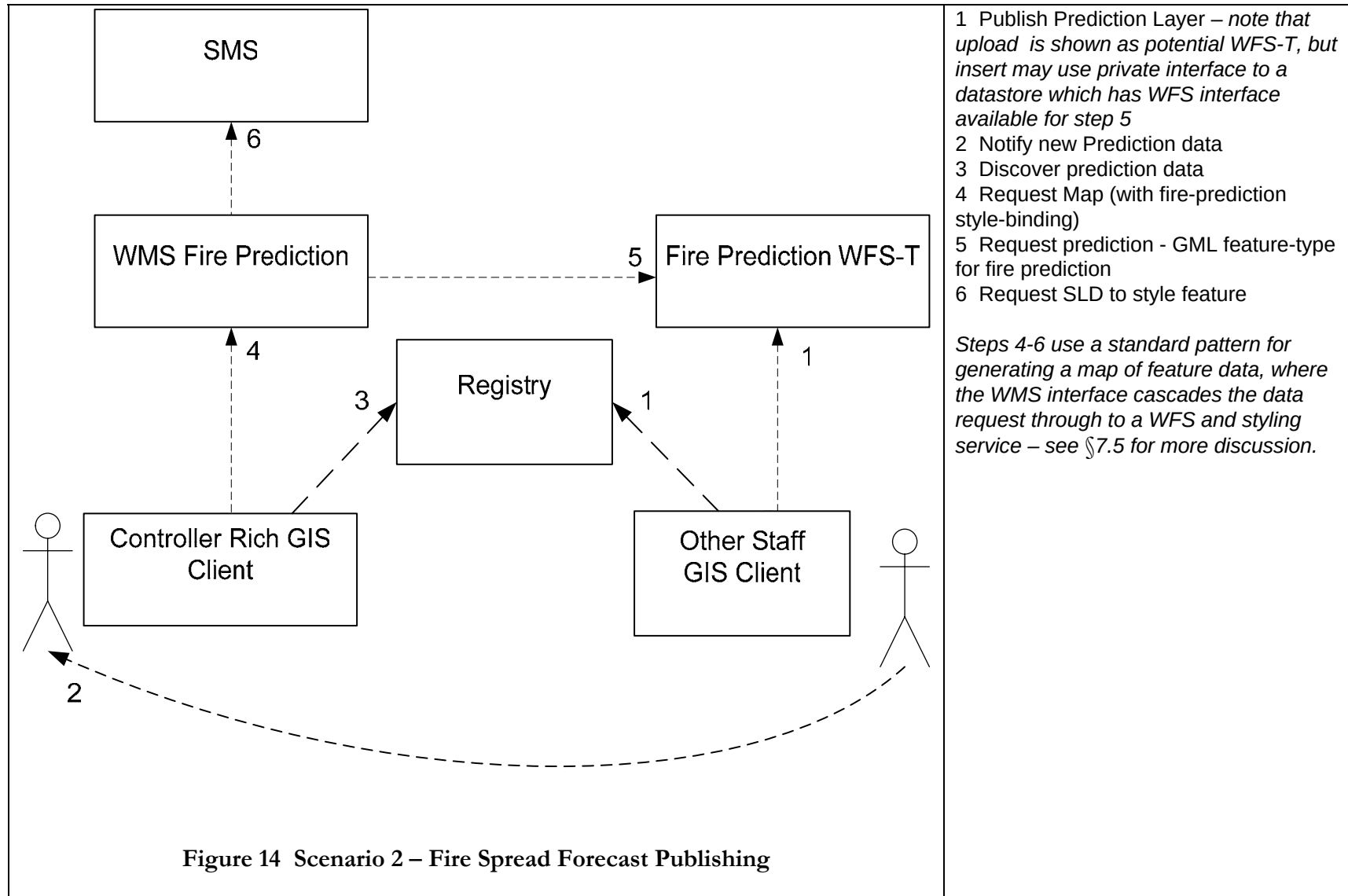
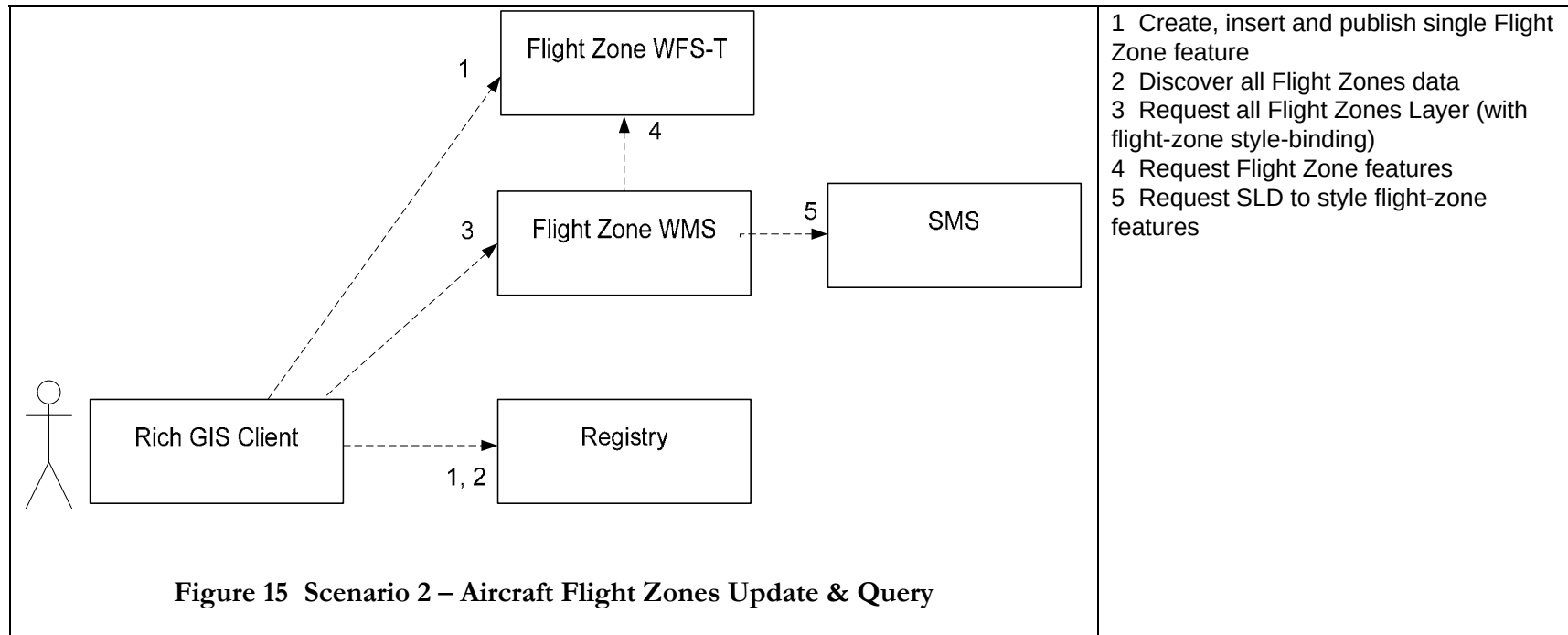
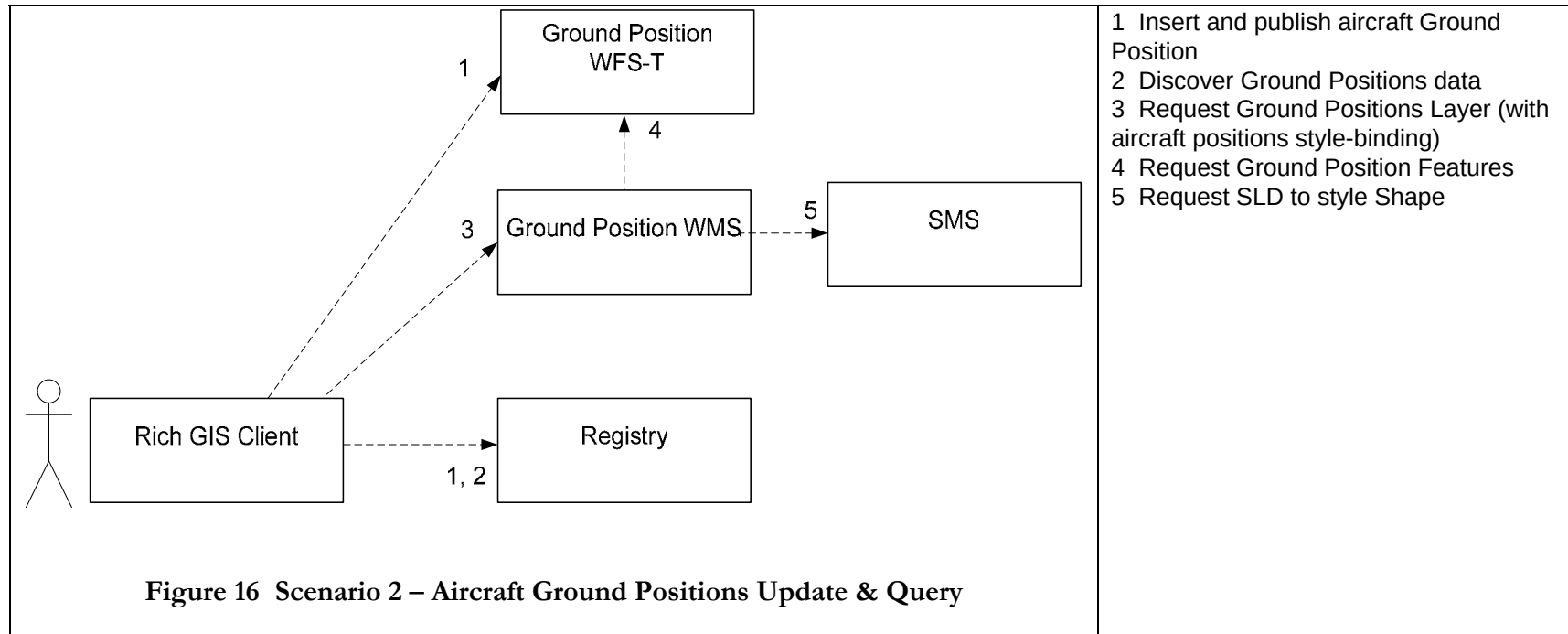
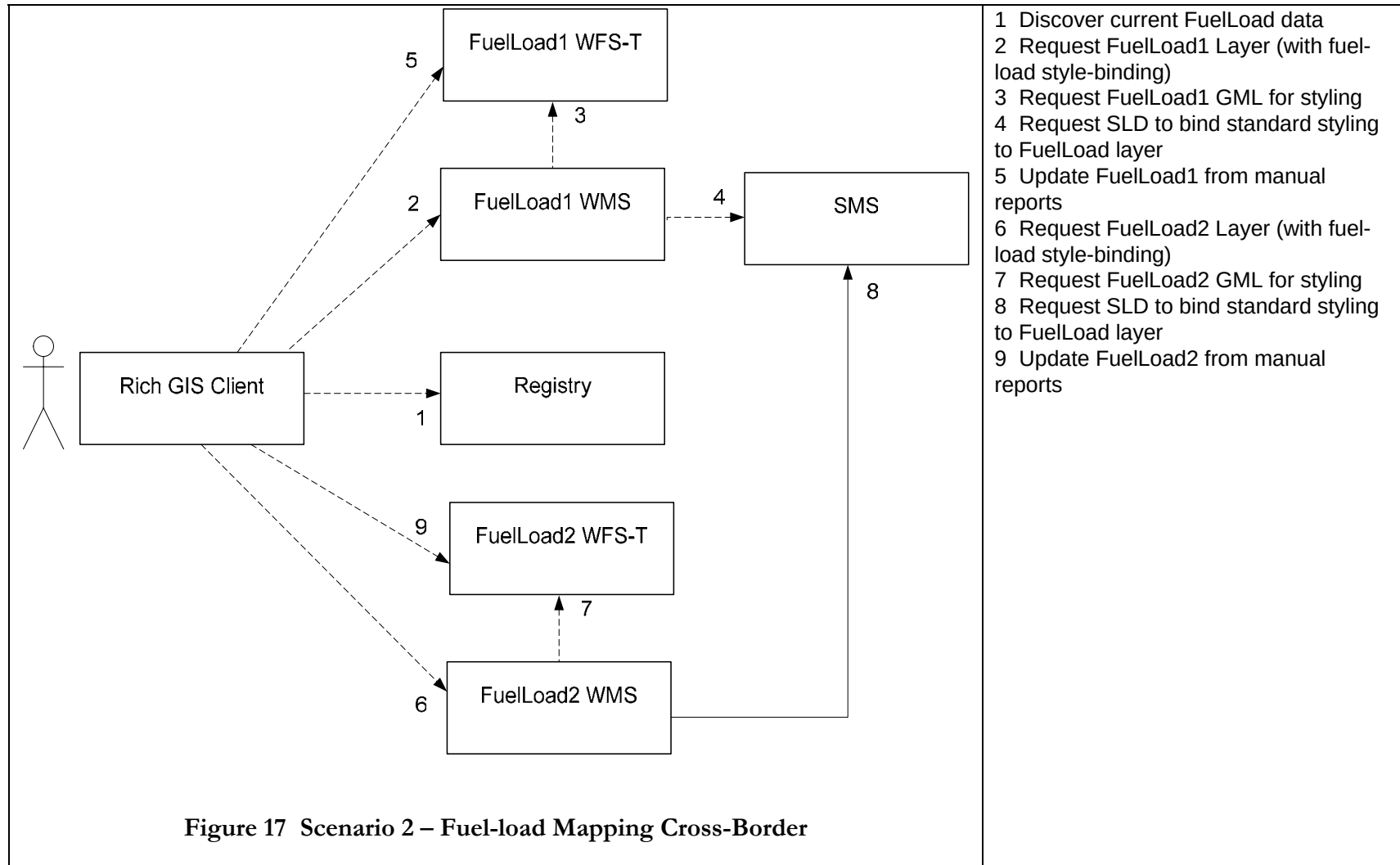


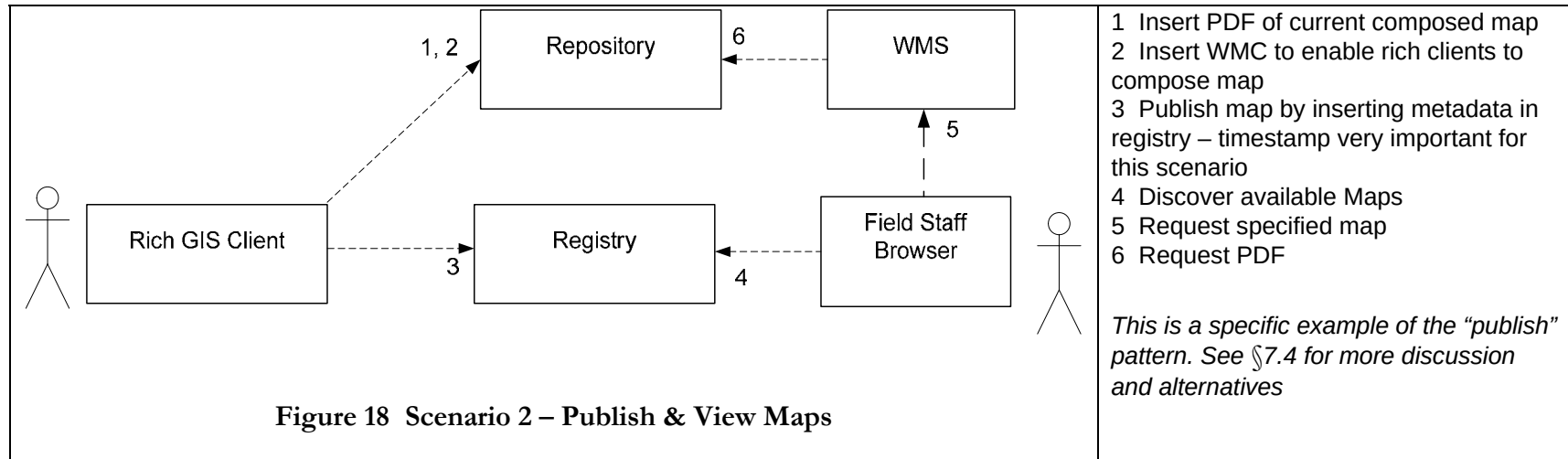
Figure 13

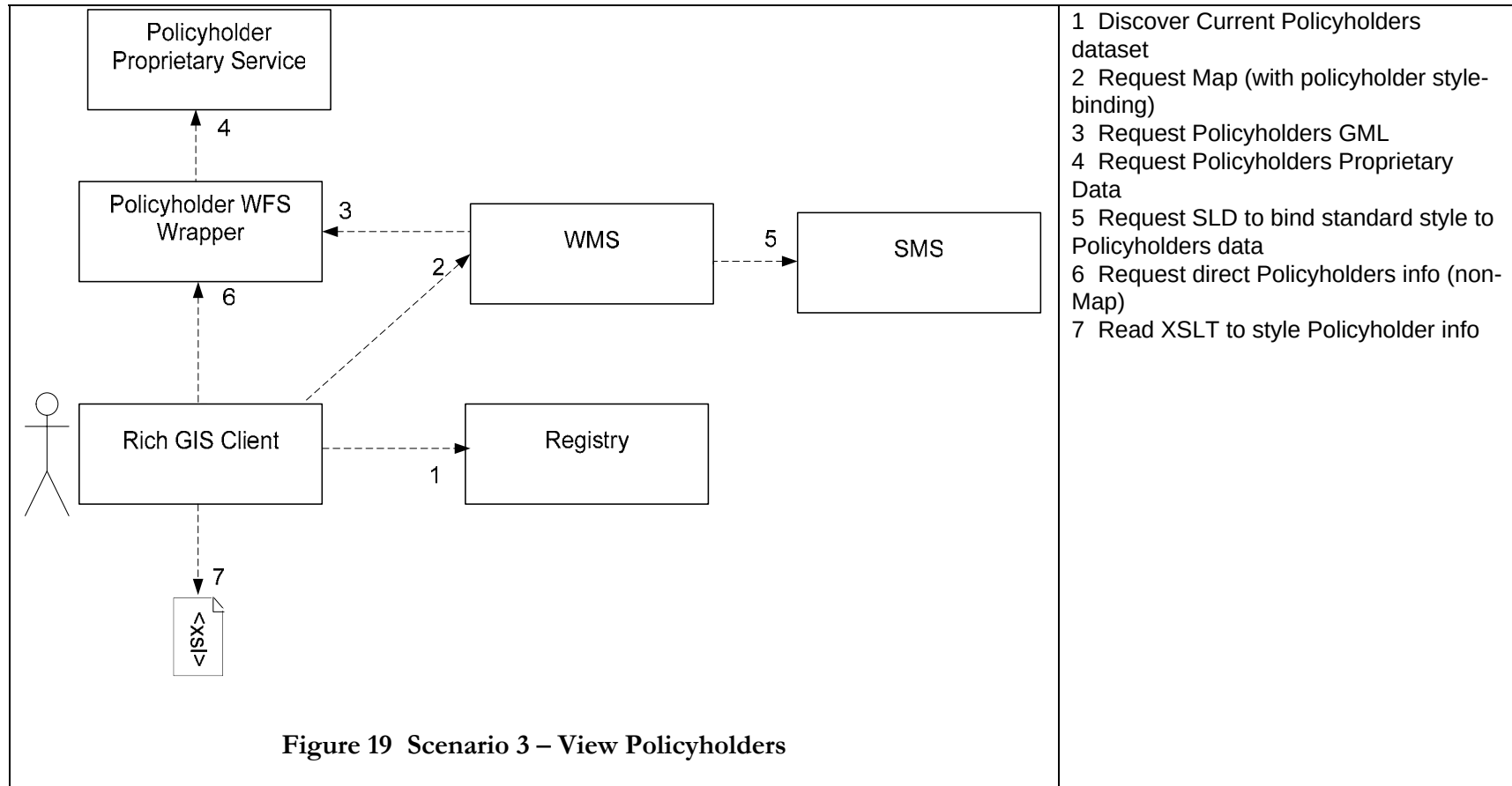


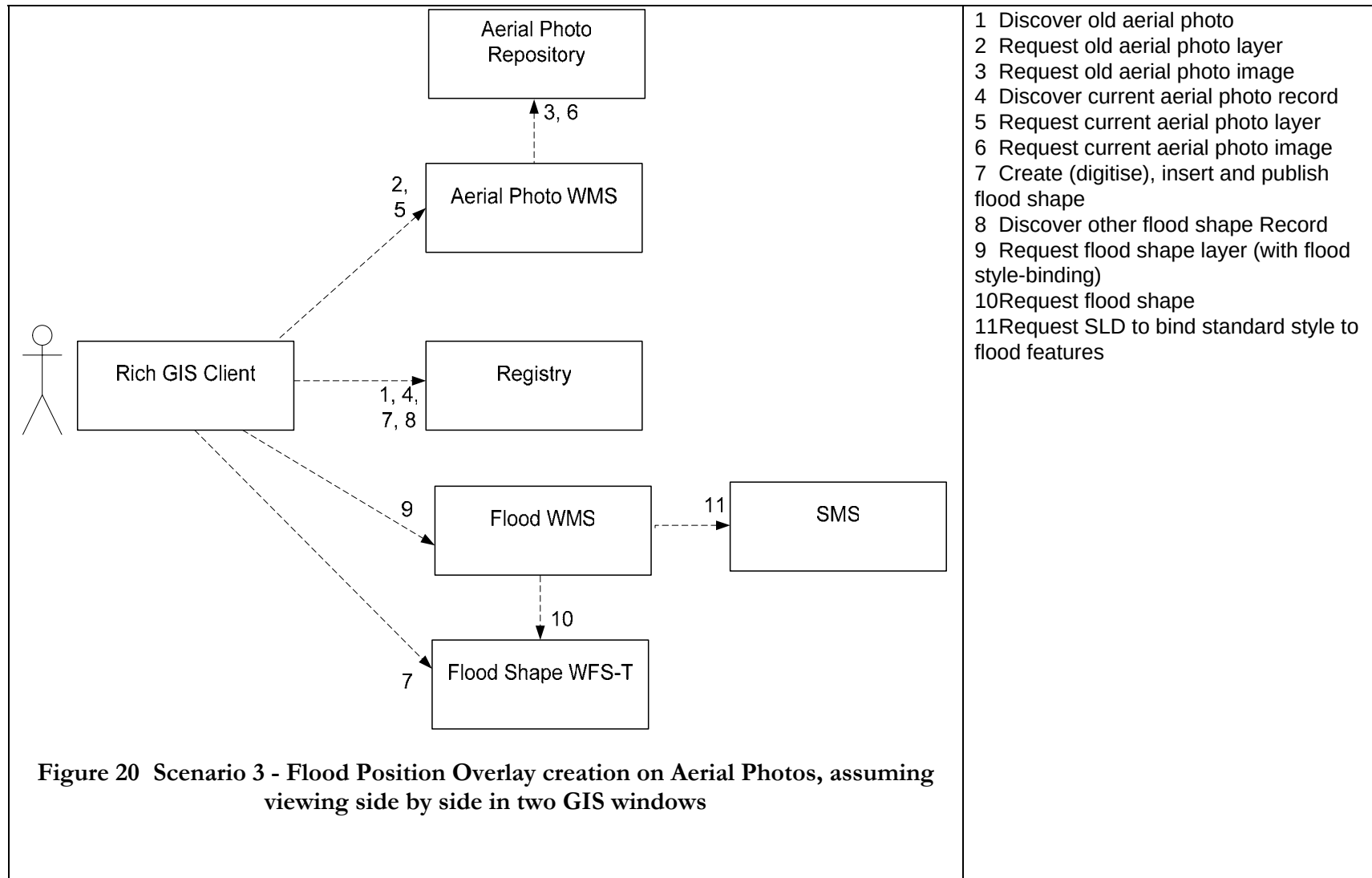












6 SIDP Application Profile

6.1 The role of content standards

The message *file-formats* are described in the interface specifications. These provide syntax and common terminology. In most situations an additional classification is encountered before actual data values are reached: these are termed **layer names**, **feature-types**, **record-types**, etc.

6.1.1 Semantic equivalence

Layer and feature-type names, and feature schemas, often reflect the storage or source schema (e.g. table structure and column names), even when provided on a public interface. However, in cases where it is necessary for a client to bind to alternative services with the same results, then a means for determining the semantic equivalence of data offerings is required. Within a map client it is often possible for the operator to infer semantics from layer/feature-type names displayed in a map legend.

However, this approach is not very robust. The general solution will use metadata. Currently the use of metadata for this purpose involves human inspection of descriptive text and keywords.

Alternatively, **content standards** can be enforced. These would require that service instances conform to standard layer and feature-naming and structure.

Thus, a WFS instance might be characterised by the feature-types to which it provides access, and a WMS by the layer names. Replacement of one data access service by another is trivial if they follow the same content standard. For example, a standard “Transport and Infrastructure WFS” should be defined in terms of the GML application schema definitions of set of feature-types that it serves, and a “Transport and Infrastructure WFS” by a standard set of layer names.

6.1.2 Standard styling

An additional benefit of standardising layer-names is that SLD [18] can be used to persist standardised style-bindings. Within the SLD syntax, each *NamedLayer* or *UserLayer* depends on a data identifier which is either

1. the “well-known” layer name (for WMS access), or
2. a feature-type name (for WFS data).

Thus, for a standard portrayal based on SLD to be useful across different service offerings, the layer or feature-type names must be shared.

6.1.3 Service configuration

In fact, for all the interfaces the content standard is an essential part of the definition of the service offering. This is generally indicated in the OWS Capabilities statement for data services, and is further specified by responses to DescribeFeature or DescribeRecord requests for WFS [21] and CS-W [3], respectively.

The large suite of different registry and library services indicated in the OGC-A Notional Architecture (V3-5 and V3-9) are primarily distinguished by their content – almost all other functions are identical between different registry and library interfaces. In these cases the N-A explicitly uses the content-model as a service-type classifier.

As indicated above, a similar approach to WFS and WMS would lead to interfaces like “Topography Feature Service”, “Vegetation Map Service”, etc. Clearly strong classification of service-types according to detailed content models is only practical for special cases dealing with very standard content types, and is unlikely to be generally scalable or useful.

6.2 Specified content standards for SIDP

In Table 3 we indicate standards and persistent resources – either existing, *proposed*, or **required**. The latter are to be acquired as part of the SIDP procurement process. These correspond to specific data transfer transactions indicated in the BON diagrams.

Following the arguments above, a suite of “standard” layer names is proposed for WMS services. As noted, this enables services to be interchangeable, and also allows standard portrayal to be recorded using SLD. Use of standard layer and feature-type names on the public interface will normally require some configuration of the WMS or WFS data service.

Table 3. Content standards and application profiles to be used in SIDP scenarios

Function	Content standard	Related event
WMS Uniform layer names: topo	Use feature type names from ICSM Harmonised model [13]: <i>InlandWaterFeature MajorWatercourse MinorWatercourse Mountain Shoreline</i>	Figure 2 /3,6
WMS Uniform layer names: infrastructure	Use feature type names from ICSM Harmonised model [13]: <i>BuildingComplex BuiltUpArea CommunicationsTower EducationFacility FireStation Pipeline Place Powerline PowerStation Railway Road Tower Utility</i>	Figure 2 /3,6
WMS Uniform layer names: incidents & events	Use incident/event type names from HSWG Symbology Reference [5]: <i>(Fire theme) HotSpot NonResidentialFire ResidentialFire SchoolFire WildFire; (HydroMeteorologicalTheme) Flood Hail LightningStrikes FireModel Smoke</i>	Figure 3 /4 Figure 8 /2 Figure 10 /7 Figure 11 /2 Figure 11 /4 Figure 14 /4 Figure 20 /9
WMS Uniform layer names: operations	Use operation type names from HSWG Symbology Reference [5]: <i>(Emergency Operation theme) EmergencyTeams; or FireSuppressionOperation (theme)¹</i>	Figure 4 /2 Figure 12 /2,8
WMS Standard layer names: weather chart	BOM NWP layers; MSL-Pressure isobars (current), MSL-Pressure isobars (forecast), WindDirection streamlines, WindSpeed isotachs, Temperature isotherms, Humidity [2] BOM Radar	Figure 6 /2
WMS Standard layer name: observation stations	BOM SYNOP stations; BOM METAR stations [2]	Figure 7 2
WMS Standard layer name: Vegetation	NDVI <i>FuelLoad</i>	Figure 9 /2 Figure 17 /2,6
WMS: Standard layer names: Aircraft	<i>Flight-zone Aircraft</i>	Figure 15 /3 Figure 16 /3

¹ The HSWG Fire Suppression Operation theme may be used for a generic fire-response feature; the specialised elements only include the fixed feature-types FireHydrant, OtherWaterSupplyLocation and FireStation.

Function	Content standard	Related event
WMS Standard layer name	<i>Map</i> (generic name for map-snapshot delivered from a WMS in static form such as as PDF)	Figure 18 /5
WMS Standard layer names - insurance	<i>Policyholders</i>	Figure 19 /2
WMS Standard layer names - Airphotos	<i>Airphoto</i>	Figure 20 /2,5
Map snapshot	PDF	Figure 18 /1,6
WFS: GML feature-types: Fire	Fire crew model Fire location model Emergency Collection Evacuation Point	Figure 4 /3,5,7,9 Figure 5 /2 Figure 10 /4,8,10 Figure 12 /3,5,7,9,11 Figure 14 /1,5
WFS: GML feature-types: Weather Observations	BOM SYNOP [2] BOM METAR [2]	Figure 7 /3,6
WFS: GML feature-types: Lightning	Lightning-strike feature-type	Figure 8 /3,6
WFS: GML feature-types: Gazetteer entry	Gazetteer entry [7]	Figure 10 /2
WFS: GML feature-types: Geocode entry	Geocode entry [8]	Figure 10 /3
WFS: GML feature-types: Aircraft movements	Flight-zone Aircraft	Figure 15 /1,4 Figure 16 /1,4
WFS: GML feature-types: Fuel-load	FuelLoad	Figure 17 /3,5,7,9
WFS: GML feature-type: policy-holder	Policy Rich feature type required for insurance scenario – including claim-status information	Figure 19 /6,3
WFS: GML feature-type:	Flood	Figure 20 /10

Function	Content standard	Related event
flood		
Style bindings	SLD [18] per scenario: Base-map (all scenarios) Power-utility Fire-threat, Fire-model, Fire-location, Fire-crew, Weather-synoptic, Weather-stations, Lightning-strikes, Smoke NDVI/vegetation, Fuel-load Flight-zone, Aircraft-position Insurance-policy Flood (See OGC Emergency Mapping Symbology testbed for sample SLD and SVG documents: http://portal.opengeospatial.org/index.php?m=projects&a=view&project_id=87&tab=2&cartifact_id=4272)	Figure 3 /6 Figure 4 /4 Figure 5 /4 Figure 7 /5 Figure 8 /5 Figure 9 /2 Figure 10 /9 Figure 12 /4,10 Figure 14 /6 Figure 15 /5 Figure 16 /5 Figure 17 /4,8 Figure 19 /5 Figure 20 /11
Style and symbol libraries	For point-located feature-types, use Emergency Symbology Reference v 2.0 where possible. New symbols required: fire-crew?, lightning, For other feature-types and layers, require standard styles for Topo, Infrastructure, Fire, based on GeoSym [12]	Figure 3 /6 Figure 4 /4 Figure 5 /4 Figure 7 /5 Figure 8 /5 Figure 9 /2 Figure 10 /9 Figure 12 /4,10 Figure 14 /6 Figure 15 /5 Figure 16 /5 Figure 17 /4,8 Figure 19 /5 Figure 20 /11
Projections, SRS	For map extents < six degrees wide, use the relevant UTM: <i>EPSG:28350 EPSG:28351 EPSG:28352 EPSG:28353 EPSG:28354 EPSG:28355 EPSG:28356 (MGA zones 50-56, GDA94)</i> ;	Figure 2 /2,5

Function	Content standard	Related event
	for broader extents use <i>Lambert Conformal Conic with standard parallels= -18, -36, central meridian 132.0, origin=(132,0), GDA94 (no EPSG code available).</i>	
WMS client initialization	A common WMC showing layers from topo and infrastructure themes, bounded by the entire operational area (for each scenario).	Figure 2 /2
WMS client specialisation	Specialised WMC showing layers relevant to scenario , bounded by the entire operational area.	Figure 2 /5
WMS composed map load	WMC	Figure 18 /2
Catalog record - datasets	OGC Application Profile for CS-W 2.0 [4], enhanced with ANZMETA controlled vocabularies for	Figure 3 /3 Figure 4 /1 Figure 5 /1 Figure 6 /1 Figure 7 /1 Figure 8 /1 Figure 9 /1 Figure 10 /4,6 Figure 11 /1,3 Figure 14 /1,3 Figure 15 /1,2 Figure 16 /1,2 Figure 18 /3,4 Figure 19 /1 Figure 20 /1,4,7,8
Catalog record - services	OGC Application Profile for CS-W 2.0	
Catalog record – other resources	Style and Symbol metadata record syntax – see Annex C from OGC 04-040 [20] WMC metadata record – use OGC Application Profile for CSW 2.0 [4], enhanced to cover additional discoverable resource types – see §6.3.2	Figure 2 /1, 4

6.3 Catalogue Application Profiles

6.3.1 Datasets and services

OGC 04-038 describes a metadata record schema compatible with CS-W 2.0. It is constructed from the components for datasets (that will be) specified by ISO 19139 (when finalized), and augmented with components for services derived from ISO 19119. It is thus a metadata record “profile” in the sense of ISO 10000 and ISO 19115 Annex C.

For use within a jurisdiction or application domain the profile must be complemented with **value domains** for many of the record elements (as reported by the CS-W **GetDomain** operation). In the context of SIDP at a minimum the ANZMETA controlled vocabularies (<http://www.ga.gov.au/anzmeta/anzmeta-1.2.html#vocab>) must be used.

6.3.2 Additional resource types

For SIDP we also require registries for Map Styles, Map Symbols, standard Style bindings (SLD), and context documents (WMC)².

For Style and Symbol catalogues, use the metadata record schema given in Annex C from Style Management Service for Emergency Mapping [20].

A “**slim**” metadata profile for WMC documents can be developed similar to the Style and Symbol metadata.

² An alternative approach to developing additional catalogue application profiles is to base them on the Catalogue Application Profile XML Schema implementation, using the relevant extension points. For example, adding additional **Resource-Identification** elements to the **smXML:_MD_Identification** substitution group. However, ad-hoc additions in this way are unlikely to scale well to all the resource types implied by V3-5 of the N-A. An alternative approach would be based on the Registry Information Model of ebXML, in which classifiers are themselves registry objects. This allows a flexible approach to ontology management. Ultimately, the deployment of a long-lasting SIDP/ASDI Registry that supports likely applications will depend on this abstraction being realised appropriately.

7 Discussion

7.1 Repositories & Libraries

OWS provide external or public access to geospatial data sources, typically using a database or GIS store on the back-end.

Maintenance of the data source is commonly through private interfaces inside the responsible organisation. Two standard OWS interfaces (optionally) support content modification (insertion, deletion and update) on a web-hosted interface³:

- WFS – in WFS-T mode, features and feature-collections may be modified in a “transaction” operation. This allows content modification across organisational boundaries, or by remote operators who have access to a slim client (typically browser-based).
- Registries – the CS-W interface [3] includes a “transaction” operation to allow a record to be added or modified. This supports the “publish” function, whereby a metadata record is added to a registry, so that subsequent users can discover a resource newly available, preferably through an OWS interface.

Within the SIDP scenarios, there is a requirement for repositories for some other information items, and complementary service-interfaces to access them. These include in particular:

- Snapshots of prepared maps, stored as PDF documents or images, to be accessed by field-crews using simple client software
- Saved map contexts, stored as WMC documents, generated by an operator using a rich client, used to load pre-configured “current-versions” of maps for use by field-crews
- SLD documents, style and symbol libraries.

There are no specific OWS interface specifications for repository service-interfaces for these information types⁴. There are two realistic options available:

1. the Catalogue Service v2 interface [3] supports access to general “Record” objects formatted as XML documents, as described in the schema exposed through the **DescribeRecord** operation. While the CS interface is primarily designed for metadata, and a “core” set of discovery-oriented elements is strongly encouraged, this is not in fact mandatory. Registry software supporting the CS v2 interface can *in principle* be configured

³ In OGC-A Notional Architecture V3-5 §5.3 and §5.4, a suite of Repository and Library services are indicated, with specific standard interfaces to these described in V3-6 and (interleaved within) V3-9. A distinction between “Repository” and “Library” is made on the basis of a broad division between geodata and workflow instances, and ancillary resources including classifiers (type libraries), spatial reference systems, and styling and symbology resources. In fact this division does not appear to be fundamental, but rather is a convenience for the discussion of a large suite of services..

⁴ Note that PDF maps could be accessed via WMS, with very restricted capabilities as described in a WMS Capabilities document similar to the one presented in §X. The repository or datastore behind this “WMS server” would simply be PDF’s in the file system. The server software might simply be the appropriate http-server (e.g. Apache) configuration that maps WMS GetMAP request to the right PDF docs!

to provide most of the registry and catalogue service interfaces described in OGC-A Notional Architecture V3-9, merely by enabling the required XML record type. Thus a CS-W v2.0 could act as a repository for both WMC and SLD documents. However, the use of CS-W as a general repository interface, configurable by specification of the record content standard, has not been broadly demonstrated and tested.

2. For the purposes of SIDP a “simple” http-server could be configured as the generic document repository interface, providing access to SLD and WMC documents “by URL”.

7.2 Registry service

A common component in all the event diagrams is the use of a registry to determine the availability and identity of specific components or services. The registry is perhaps the key infrastructure item in a loosely-coupled SOA, in which applications are assembled as needed by composition and choreography of sets of specific service instances. The first service needed by any client – thick, thin or chubby – is a means for selecting datasets based on matching their semantics to the scenario requirements, the service components of various types, and supporting components used for configuration and transformation, particularly in portrayal.

All datasets, feature-types, services, style bindings, context documents that are used in SIDP should be catalogued in the SIDP registry, which may serve as either prototype or core of a more general registry deployment within ASDI.

This does *not* (necessarily) involve the development of new applications software, but does require detailed attention to record schemata, application profiles, the relationships between objects at different metalevels (e.g. datasets vs feature-types) and language bindings (e.g. GML vs shapefile for the same feature-type) and how these may be used to classify other resources (e.g. feature-types for data-access services), and the governance models used to configure the registry installation.

While it might be possible to generate small extensions to the available profiles to accommodate the information types in the scenarios, we can be confident that these only represent the thin-end of the wedge in terms of likely description and classification requirements as more scenarios are explored. The range of registry services indicated in the OGC-A Notional Architecture indicate some of the possible scope, and within even each of this set of registry types, there are likely to be a variety of record types as well as classifiers.

However, the approach available for classification of resources in the CS Profile [4] is quite restricted, and does not effectively leverage the scalable ebXML Registry Information Model referred to in the OGC-A Notional Architecture (OGC-A Notional Architecture V3-5 §5.5 – component 5.5.12). The *strength* of the RIM model is that the classification schemes are themselves merely other registry items. The *challenge* is establishing appropriate high-level starting points, and the necessary governance model which enables a registry to accommodate all the resource types likely to be used within the application domain.

It is recommended that **detailed specification of a registry service** for SIDP (and ASDI) that is properly scalable and capable be the topic of a **distinct work-item**.

7.3 Setup and initialisation

There is a certain amount of setup which must be accomplished even to do a walkthrough of the demonstrator. This mimics what would be accomplished in a Purchasing Phase of negotiating SLAs, but also includes other application and document creation tasks.

Where the narrative refers to preconfigured maps these may be described using WMC documents. The bounding box of a WMC used for initialisation will cover the complete area of responsibility of the operator, so an early step by the operator will be to zoom to the area of interest.

Thus, at a minimum, the setup phase should include

- creating **WMC documents** which describe maps based on layers that satisfy the scenario starting point, and thus serve to initialise the client, and
- binding the client to required **registries or registry stubs**

In practice the client application might be preconfigured for the scenario with the initial context fixed, but this should still be formally documented as a WMC.

7.4 Publishing Maps

The “classic” publish-find-bind pattern of a loosely-coupled service-oriented-architecture may be explicitly visible to the operator and participants in one common sub-scenario: issuing maps or map products (Figure 21).

Publication requires a metadata record describing the resource to be lodged in the necessary registry. The map resource itself can be in form of

- (i) A snapshot in a static format such as PDF. The map must be inserted in a repository, which may have a OWS interface such as WMS, or may simply be a file on a simple web-server. The metadata is a standard “dataset” record, with the URL for the map recorded as the online-resource. The content in a snapshot is timestamped by its compilation time.
- (ii) a new **layer** or **feature collection** added to a datastore. To be useful in a web-mapping context, the datastore must have a WMS interface available, which will be identified in the metadata record. The metadata is a standard “dataset” record. The content in a layer is timestamped by its creation time.
- (iii) a composite map that can be described in a single **GETMAP** request. The metadata is a variant dataset record with the GETMAP URL recorded as the online-resource. The content in a map obtained in response to a GETMAP request is timestamped by the request time, and may use layer(s) that may have been updated since the GETMAP request was published by creation of the registry record.
- (iv) a composite map whose description is more elaborate than can be encoded in a GETMAP URL, and thus needs a **WMC document** that may be picked up by a WMS client that supports WMC. This requires a WMC repository (possibly just a simple web-server). The metadata is a variant dataset record with the WMC URL recorded as the online-resource. The content in a map obtained in response to a set of GETMAP requests implied by LAYER elements in a WMC is timestamped by the request time, and may use layer(s)

that may have been updated since the WMC was added to the WMC repository, and published by creation of the registry record.

Several registries may be involved, with different levels of access reflecting publication for different audiences. For example, a “work-in-progress” might be “published” for internal or restricted access by creating a record in a private registry. A resource that has passed through the workflow and necessary approval procedure for external access shall have a record created in a more widely accessible or public registry.

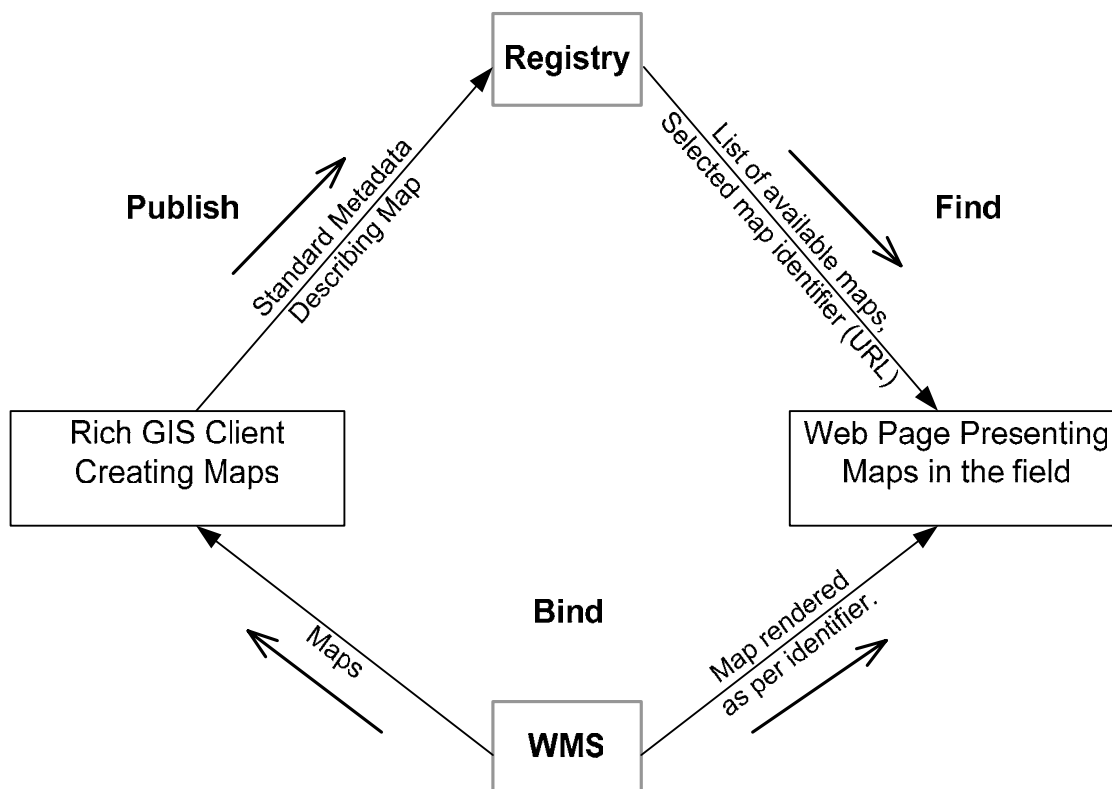


Figure 21 Map Publishing Overview

7.5 Map portrayal and styling

Most of the spatial analysis operations appearing in the scenarios require visual inspection of the data, presented as *maps*. Map generation requires the conversion of information into visual form through

- Selection of the information to be portrayed, by theme and location
- Projection of the spatial representation of the data into the required 2-D window
- Generating a graphical representation using styles and symbols that communicate effectively to the viewer

Each of these three steps requires application of information profiles⁵.

⁵ There are relatively well-known conventions for some of these aspects: e.g. detailed maps are usually presented using TM projections in standardised zones; water bodies are usually coloured blue. However, other aspects are not standardised by convention: e.g. layer names or feature-type names vary widely between providers; projections for broader scale maps are much more variable. Furthermore, lack of standardisation, in some areas, does not necessarily create difficulties for the user.

Within the OWS interfaces, the issues listed above correspond to the following information items:

- Information semantics are indicated
 - for WMS using **layer names**
 - for WFS using GML encoded **feature types**
- Coordinate reference systems are indicated by the value of **SRS** or **srsName** elements and attributes within XML encodings for WMS and WFS/GML
- Portrayal is defined
 - in WMS by **named styles** pre-configured on a layer-by-layer basis,
 - for conformant WMS or WFS, by links to an external source, or explicit description, in a **Styled Layer Description** (SLD) document
 - SLD also binds a style to a layer or feature-type.

Each of these mechanisms may be used as a means of standardisation or persistence.

Each audience, including the primary operator, should have a standard view including styling. so operators or users can be replaced without affecting interpretation. Fixed styling can be enforced by

1. WMS's used in SIDP are configured with fixed styles (indicated in the Capabilities) that follow standard bindings to the layer semantics
2. WMS's used in SIDP
 - a. must support SLD,
 - b. are configured with standard layer names (indicated in the Capabilities)
 - c. standard SLDs are provided to be used by styling agents including clients in SIDP scenarios.

Finally

- specific combinations of styled map themes in a specific window and projection may be recorded for a WMS client in a **Web Map Context** (WMC) document

The last are used particularly for client initialisation (see 7.1 above), but may also be used for publication of data products where it is required to produce the “current version” by combination of specified layers at bind-time (see 7.4). Note that WMC includes support for styling using SLD. In principle this provides a mechanism for including WFS feature types in a persistent map description. However, the current version of WMC does not allow this properly on syntactic grounds, so where this method is used **for WFS data an extension to WMC** is required.

Following the OGC-A Notional Architecture V3-8, the WMS interface is the primary means for delivering map data constructed for visual consumption. This appears as the most common interface within the component diagrams. However, in several cases the content being portrayed is obtained from a secondary feature-store. Consistent with the

For example, the shape of many features together with hints provided by layer names emitted from the data source and listed in a legend often allow the semantics to be inferred. And when the information is being provided by a single map service, varying projections between layers are not usually a problem. However, when information is being combined from multiple sources, then it is critical that a common profile is used. When this is not by convention, then standard application profiles must be explicitly specified.

general approach taken in specifying the architecture, we indicate that the WMS shall obtain feature data using a WFS interface. Detailed subsidiary architectures to accomplish this may follow the patterns described in Annex G of Style management Services for Emergency Mapping Symbolology .[20]

7.6 Content creation

During the progress of a scenario, new geospatial data may be introduced through

- transformation of existing content, e.g. classification of an image, conversion of a collection of features into a new layer, merging of two or more layers within a specific bounding box. This method inherits spatial properties from the predecessor data
- addition of data simple (point-located) geometry through a text or form interface.
- Addition of data with simple (point) or complex geometry (line, curve, polygon or area) through picking or digitizing in a rich client. This may use a java component within a browser-hosted client, or may accomplished using a web-enabled desktop client.

However, note that in general the provision of web-hosted access to information services should not be intrusive on the normal data management procedures of a custodian organisation. The web-service interfaces should be seen as value-adding by enabling use of information resources by a wider group of users, through a more flexible set of client applications.

7.7 Non-graphical data

In certain operations, detailed information is required by the operator that goes beyond what may be reasonable presented in a map. For example, detailed weather observations or predictions typically involve many parameters at a location; contact information for a crew or property requires structured addresses or identifiers. In these cases the WFS interface is used to request detailed information concerning specified features, which may include multiple attributes or properties for a feature instance, and complex values for particular attributes.

In most cases selection of the object of interest is done visually, through a map interface, but then the report of the details of the object will be presented in textual or tabulated form. This final presentation will usually require transformation of information supplied as a GML instance into HTML or some other presentation format. This transformation can be specified using an XSLT script, selected by the client application, with the actual processing in a client application mostly likely based on standard web-browser support for XSLT transformation.

7.8 WMS vs WFS

In a number of positions within the scenarios, implementation via either WFs or WMS is possible. In general WMS is the more mature technology, and *providing the content standards specified here are followed*, will enable a simpler architecture. A few considerations would lead

to the requirement to deploy WFS services (and development of the required GML Application Schema): If either

Update/insertion

Access to complex attribute/property data

Are required, then the WFS interface is needed. Otherwise, only a WMS interface need be exposed (even if it cascades invisibly to a WFS).

7.9 Authentication and security

Authentication shall be performed by http realm-based authorization using SSL encryption to guard communications.

It is important to note that only standards-based wrapper services are used for web-hosted access to proprietary systems. These WFS and WMS wrappers can generally provide secure portals to the proprietary systems without having to expose the network or service behind them.

Table 4 shows the minimum set of points at which access control can be separately specified.

Table 4. Authentication Interfaces

User	Scenario	Event Diagram	Accesses	Implications
Web browser running Control Room Polling Application	1	Fig 3	Control Room Polling Application	The polling application must have access to a registry and the Firewatch WMS Wrapper. This could well be within a single server. The browser client only needs access to the polling application, not to any other parts of the system.

User	Scenario	Event Diagram	Accesses	Implications
Main GIS Client	All	Most	Everything	The Main GIS Client must be accepted by multiple servers which may be disjoint authentication domains. As part of the demonstrator setup, we must make sure it has been authorised in each realm. The client must be capable of remembering multiple passwords for different realms. The client must also be capable of querying the service to establish which authentication domains it is willing to accept. Services may accept authentication from a sufficiently trusted different domain rather than requiring the client to authenticate directly in their domain.
Other Staff GIS Client	2	Fig 10, 13	Everything	The Geocoder and Gazetteer services in Fig 10 are coupled to the rest of the system only via the query from the Other Staff client so could be outside the main secure area, with their own security zone.
Field Staff Browser	2	Fig 17	Map Publishing Registry & Repository	The registry must have a public read-only catalog service. This is not the same as the service used to publish metadata. Similarly, there's a read-only service to access the PDF maps.

User	Scenario	Event Diagram	Accesses	Implications
GPS Proprietary Wrapper	1	Fig 4	Crew WFS-T	Alternatives: 1) An external interface for transactional updates on this WFS-T is separately authorized., or 2) This external service has access to all data.

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9

